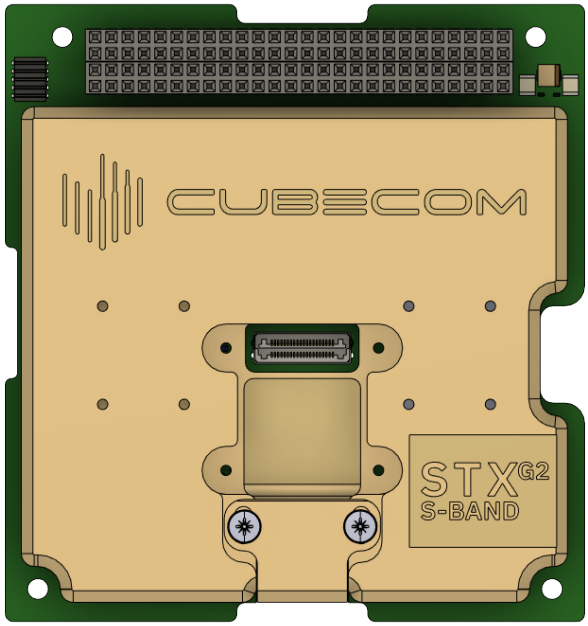
VestaLink S25 (STXG2)

USER MANUAL

UID:	USM-01-10022-A8
Template:	TPL-30-00001-B1
Product UID	01-10022
Product Name	VestaLink S25 (STXG2)
Product Description	Commercial S-Band Nano Satellite Transmitter (25 Mbps; 2.2-2.29 GHz)



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4 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Term	Description / Definition
APSK	Amplitude and phase-shift keying
ASM	Attached sync marker
BAT	Battery
BB	Baseband
CAN	Controller area network
CCSDS	Consultative Committee for Space Data Systems
Compliment	Refers to the inverted output of a differential driver
Conf	Configuration
CRC	Cyclic redundancy check
Ctrl	Control
DAC	Digital to analog converter
dBm	Decibels relative to one milliwatt
DDR	Double data rate
DFL	Data Field Length
DMA	Direct memory access
DNC	Do not care
DNP	Do not populate
DVB-S2	Digital Video Broadcasting – Satellite 2
EEP	Error end of packet
eMMC	embedded MultiMediaCard
EOF	End of file
EOP	End of packet
ETSI	European Telecommunications Standards Institute
EVM	Error Vector Magnitude
FECFRAME	Forward error correction frame
FIFO	First-in, first-out
FPGA	Field-programmable gate array
FW	Firmware
GaN	Gallium Nitride
GHz	Gigahertz
HW	Hardware
I/O	Input and Output
I2C	Inter-integrated circuit
ICD	Interface control document
ID	Identification
IFC	In-flight configurable
OID	Only idle data
kHz	Kilohertz

Term	Description / Definition
LDPC	Low-density parity-check code
LEO	Low earth orbit
LFSR	linear feedback shift register
LSB	Least significant byte
LVDS	Low voltage differential signaling
Mbps	Megabits per second
MC	Master Channel
MHz	Megahertz
MODCOD	Modulation and Coding
MSB	Most significant byte
Msps	Mega samples per second
N/D	Not detected
OBC	On-board computer
OCF	Operational Control Field
OOB	Out of bounds
OSI	Open Systems Interconnection
PA	Power amplifier
PC	Personal Computer
PCB	Printed circuit board
PID	Proportional–integral–derivative
PLFRAME	Physical layer frame
PPS	Pulse per second
PRBS7	Pseudorandom binary sequence (7-bit)
PSK	Phase-shift keying
QPSK	Quadrature Phase Shift Keying
QSPI	Quad SPI
R	Read
RF	Radio frequency
RMS	Root mean square
RRC	Root-raised-cosine
RW	Read/Write
SCID	Spacecraft ID
SDR	Single data rate
SERDES	Serializer/Deserializer is a highspeed serial communication standard
SFCG	Space Frequency Coordination Group
SMTF	Synch-Marked Transfer Frame
SPI	Serial Peripheral Interface
SpW	SpaceWire
SW	Software
TBC	To be confirmed
TBD	To be determined

Term	Description / Definition
TF	Transfer frame
TFPH	Transfer frame primary header
TFSH	Transfer frame secondary header
TM	Telemetry
TRUE	Refers to the non-inverted output of a differential driver
TM/TC	Telemetry / Telecommand
UART	Universal asynchronous receiver transmitter
USM	User manual
VCID	Virtual Channel ID
VCM	Variable coded modulation
W	Write
w/	with
w/o	without

5 DOCUMENT REFERENCES

Table 5-1: Document References

	Standard reference	Description
1	ISO/FDIS 17981	Space systems — Cube satellite (CubeSat) interface
2	CCSDS 132.0-B-3	TM SPACE DATA LINK PROTOCOL
3	CCSDS 131.0-B-5	TM SYNCHRONIZATION AND CHANNEL CODING
4	CCSDS 131.3-B-2	CCSDS SPACE LINK PROTOCOLS OVER ETSI DVB-S2 STANDARD
5	ETSI EN 302 307-1 V1.4.1 (2014-11)	Digital Video Broadcasting (DVB); Second generation framing structure, channel coding and modulation systems for Broadcasting, Interactive Services, News Gathering and other broadband satellite applications, Part 1: DVB-S2

6 INTRODUCTION AND SCOPE

6.1 IDENTIFICATION

The VestaLink S25 S-Band transmitter is defined as CUBECOM part number 01-10022. This document is identified as USM-01-10022-A9.

This document is the user manual for the VestaLink S25 and describes the operation, performance and register map thereof. Refer to the ICD document (ICD-01-10022) for details on the electrical and mechanical interfaces.

This product was previously known as the “STXG2” and has been renamed to “VestaLink S25”. The product remains unchanged, with only the product name being updated.

6.2 INTRODUCTION

The VestaLink S25 is a high-performance S-Band transmitter for use in nano satellites. It can support a maximum encoded data rate of up to 25 Mbps and operates over the frequency range of 2.2 GHz to 2.29 GHz, making it suitable for S-Band communication systems. The maximum symbol rate of the transmitter is 5 Msps and it supports both CCSDS [3] and DVB-S2 [5] (MODCODs from 1 to 28) encoding. This user manual is intended to provide instructions and information regarding the configuration and operation of the VestaLink S25 transmitter. It is essential that the user reads and understands the contents of this manual before operating the transmitter.

Figure 6-1 below shows a perspective view of the VestaLink S25.

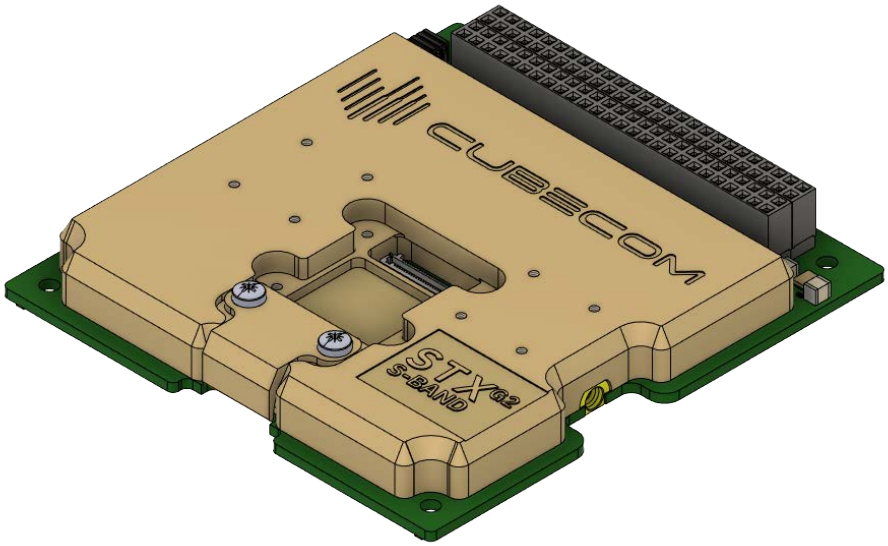


Figure 6-1: VestaLink S25 perspective view.

6.3 COMPARISON TO THE AstraLink X100 (XTX GEN1)

The VestaLink S25 is the next generation S-Band version of the AstraLink X100 (XTX Gen1). A summary of the high-level differences between the two generations of the radio is summarized in Table 6-1 below.

Table 6-1: Specification Comparison between the AstraLink X100 (XTX) & the VestaLink S25 .

Parameter	XTX	VestaLink S25
Frequency	8.025 GHz to 8.4 GHz	2.2 GHz to 2.29 GHz
Encoding	DVB-S2 [5] or CCSDS [3]	DVB-S2 [5] or CCSDS [3]
Modulation	QPSK / 8-PSK / 16-APSK	QPSK / 8-PSK /16-APSK / 32-APSK
Maximum Symbol Rate	25 Msps	5 Msps
Maximum Encoded Data Rate	100 Mbps	25 Mbps
Internal Memory	None	32 GB
Tx Power	0.5 - 2 W	0.5 - 2 W
Power Consumption	< 15 W	< 12 W
Power Supply Range	7 – 20 V	7 – 20 V
Operating Temperature	-20 – 60 °C	-20 – 70 °C
Dimensions	96x90x13 mm	96x90x13 mm
Mass	120 g	165 g
Lead Time	8 - 12 Weeks	8 - 12 Weeks

7 TRANSMITTER OVERVIEW

The VestaLink S25 is developed to fit in a standard PC104 stack as defined by the Pumpkin standard (ISO/FDIS 17981) [1].

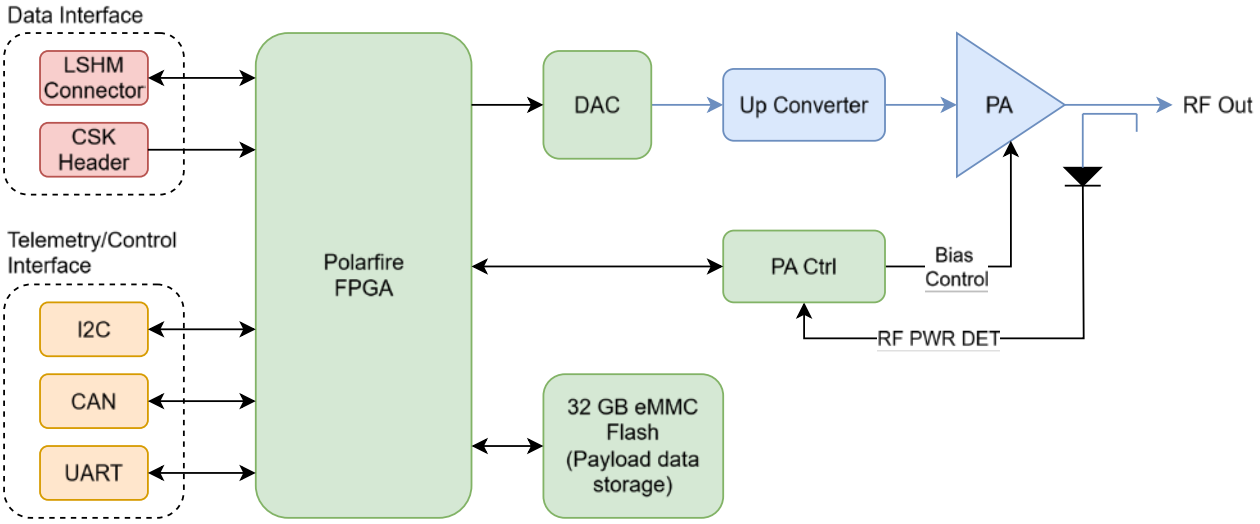


Figure 7-1: VestaLink S25 interface overview.

The design of the system can be divided into several distinct components which include TM/TC, data interfaces, the FPGA, RF logic, and power supplies. The TM/TC interfaces currently supported are I^2C, CAN (dual/redundant CAN option available) and UART. The data interface consists of a SPI interface implemented on the PC104 header and LVDS pairs that can be configured into various high speed data interfaces on the LSHM connector.

Sattelite bus connectivity is provided via the PC104 connector. Power and the TM/TC interfaces are supplied through this connector, along with a few control I/O's. The VestaLink S25 is powered from its V_{BAT} pin on the PC104 connector and has an input voltage range of 7 V - 20 V. The VestaLink S25 may be powered down externally via an optional Enable line located on the PC104 header. All inputs and outputs through the PC104 connector are isolated from the FPGA with protection buffers to prevent dead logic from being back powered through the connector(s) when the board is powered off. The connections through the LSHM connector are not isolated from the FPGA to facilitate higher speed data links.

Furthermore, the VestaLink S25 implements a highly linear GaN power amplifier to produce an RF output of up to 2 W. The power amplifier has a power detector integrated for dynamic RF power control and monitoring. The FPGA is controlled by a firmware-based processor. All control and monitoring functions are implemented via this processor. Critical safety related monitoring and protection such as over current and over temperature shutdown of the PA is implemented in hardware. The high-speed data path and encoding are implemented entirely in the FPGA's firmware.

The VestaLink S25 features integrated DVB-S2 and CCSDS cores for encoding and modulation, with final output upconverted to S-Band. The VestaLink S25 supports DVB-S2 MODCODs 1–28, encompassing QPSK, 8PSK, 16APSK, and 32APSK constellations in full compliance with the DVB-S2 standard. The maximum symbol rate for this transmitter is 5 Msps, which at 32-APSK (DVB-S2) modulates 5 bits per symbol to generate a modulated signal with a maximum encoded data rate of 25 Mbps. The transmitter covers an output frequency range of 2.2 GHz to 2.29 GHz.

The VestaLink S25 emissions mask conforms to SFCG 21-2R4 recommendations. The center frequency of operation is in-flight configurable in 1 MHz steps. The RF output power is also adjustable in-flight from 27-33 (FMs) / 14-20 (Flatsats) dBm in 1 dB steps.

7.1 TM TRANSFER FRAME ENCAPSULATION

The VestaLink S25 implements the CCSDS TM space data link protocol as defined by the CCSDS RECOMMENDED STANDARD FOR TM SPACE DATA LINK PROTOCOL [2].

The data from the selected data source or flash memory is encapsulated with a Transfer Frame Primary Header (TFPH), Transfer Frame Secondary Header (TFSH) and CRC (if enabled) at the end as defined by the CCSDS standard [2] for transfer frames.

The byte breakdown of the complete transfer frame is shown in Table 7-1. A detailed breakdown of the TFPH, TFSH and CRC is given in the subsections that follow.

Table 7-1: Transfer frame structure

TM Transfer Frame			
Transfer Frame Primary Header (TFPH)	Transfer Frame Secondary Header (optional, TFSH)	Transfer frame data field (Payload data)	Transfer frame trailer Frame Error Control Field (optional, CRC-16)
6 bytes	4 bytes	200 - 16380 bytes	2 bytes

7.1.1 TRANSFER FRAME PRIMARY HEADER

Table 7-2 below shows the breakdown of the TFSH. Values indicated with as (IFC) are in-flight configurable and are accessible through the register map (see Section 9).

Table 7-2: Transfer Frame Primary Header structure (6 bytes)

TM Transfer Frame						
Master Channel ID		Virtual Channel ID	OCF Flag	Master Channel Frame Count	Virtual channel frame count	Transfer Frame Data Data Field Status
Transfer Frame Version Number	Spacecraft ID					
b”00” 2 bits	(IFC) 10 bits	(IFC) 3 bits	b”0” 1 bit			
2 bytes				1 byte	1 byte	2 bytes

The following list further details the primary header structure described in Table 7-2:

- The **Master Channel ID** consists of the transfer frame version and the Spacecraft ID.
    - The **Transfer Frame Version Number** is “00” bits for the referenced protocol standard.
    - The **Spacecraft ID** is used to identify the spacecraft which is associated with the data contained in the Transfer Frame.
- The **Virtual Channel ID** is used to separate data into different channels.
- The **OCF flag** is 0 and indicates that the Operational Control Field is not present.
- **Master Channel Frame Count** is an ‘8-bit modulo 256’ count of the master frames generated. This counter increments with every generated transfer frame, including idle frames.
- The **Virtual Channel Frame Count** is an ‘8-bit modulo 256’ count used to count virtual frames transmitted for a specific virtual channel. There is a counter for every virtual channel and is only incremented for the associated virtual channel.
    - IDLE frames are always associated with virtual channel 7.
    - The virtual channel for the payload data is in-flight configurable.
    - The virtual channel for the retransmission data is also in-flight configurable and may be different than the payload data virtual channel (DVB-S2 only).
- The **Transfer Frame Data Field Status** identifies valid data and idle packets (Table 7-3).
    - For valid data, this field is 0x9800.
    - For Only Idle Data (OID), this field is 0x9FFE. This field is used to discard/filter out the idle data in the ground segment processing.

Table 7-3 below shows the breakdown of the Transfer Frame Data Field Status field of the primary header. Values indicated as (ENC) is dependent on the selected encoding & modulation scheme.

Table 7-3: Transfer Frame Data Field Status structure (2 bytes)

Transfer Frame Data Field Status				
Transfer Frame Secondary Header Flag	Synchronization Flag	Packet Order Flag	Segment Length ID	First Header Pointer
(ENC) 1 bit	b"0" 1 bit	b"0" 1 bit	b"11" 2 bits	Packet type dependent 11 bits
2 bytes				

The following list further details the Transfer Frame Data Field Status structure described in Table 7-3:

- The **Transfer Frame Secondary Header Flag** indicates if the Secondary Header is present (1) or not (0) in the VestaLink S25's implementation of the CCSDS header structure.
    - When using DVB-S2 encoding & modulation, this flag is set to 1 (TFSH present).
    - When using CCSDS encoding & modulation, this flag is set to 0 (TFSH not present).
- The **Synchronization Flag** is 0 to indicate the inserted data is octet-synchronized and forward-ordered Packets or Idle Data are inserted.
- The **Packet Order Flag** is 0 and reserved for future use.
- The **Segment Length ID** is set to "11" as defined by the standard.
- The **First Header Pointer** contains the position of the start of the data packet in the Transfer Frame Data Field. Each Transfer Frame of the VestaLink S25 contains a complete packet. Each packet therefore starts at the start of the Transfer Frame Data Field. Reading from a section of flash memory that has been corrupted or that has not been previously written will read as undefined data. This status field can be used to determine if the data in the packet is Only Idle Data (OID) or user data.
    - OID is indicated with b"11111111110".
    - User data is indicated with b'0000000000'.
    - Undefined data is indicated with b'1111111111' (DVB-S2 only).

7.1.2 TRANSFER FRAME SECONDARY HEADER

Table 7-4 below shows the breakdown of the TFSH.

Table 7-4: Transfer Frame Secondary Header structure (4 bytes).

Transfer Frame Secondary Header (TFSH)			
TFSH ID		**TFSH Data Field**	
Transfer Frame Secondary Header Version Number	Transfer Frame Secondary Header Length	EOF Flag	Packet ID
b”00” 2 bits	b”000011” 6 bits	1 bit	23 bits
1 byte		3 bytes	

The Transfer Frame Secondary Header consists of the end of file (EOF) flag and the payload data Packet ID to uniquely identify a block of data in a transmission. The Packet ID is used by the ground segment to verify the data integrity and if there is corrupted or missing data, it can be used as a reference to request retransmission of the implicated blocks of data.

It is important to note the following about the transfer frame secondary header structure:

- The **TFSH ID** is constant with a value of 0x03. The header version number is 0 and the TFSH length is the total length minus one which is 3.
- The first bit in the TFSH data field is an **EOF flag** that the VestaLink S25 generates at the end of a write cycle or if the data source generates an EOF character.
- When framing data from on-board flash memory, the **Packet ID** represents the memory index from which the packet was retrieved. The Packet ID is represented by the the Frame ID shown in Figure 7-7.
- When the framer receives payload data from a source or test pattern generator, the Packet ID serves as a 23-bit incrementing counter to ensure data continuity.

7.1.3 TRANSFER FRAME ERROR CONTROL

The transfer frame is completed with two CRC-16 bytes (if enabled). The CRC-16 is created using the polynomial G(X) shown in the equation below:

$$G(X) = X^{16} + X^{12} + X^5 + 1$$

The CRC-16 logic diagram is shown in Figure 7-2. The shift register is initialized with '1's at the start of each transfer frame. The transfer frame is shifted through the shift register (Excluding the ASM). The transfer frame is sent to the DVB-S2 or CCSDS core to be encoded and transmitted while it is shifted through CRC-16 shift register. At the end of the transfer frame the 2 bytes of the CRC-16 are shifted out and appended to the transfer frame.

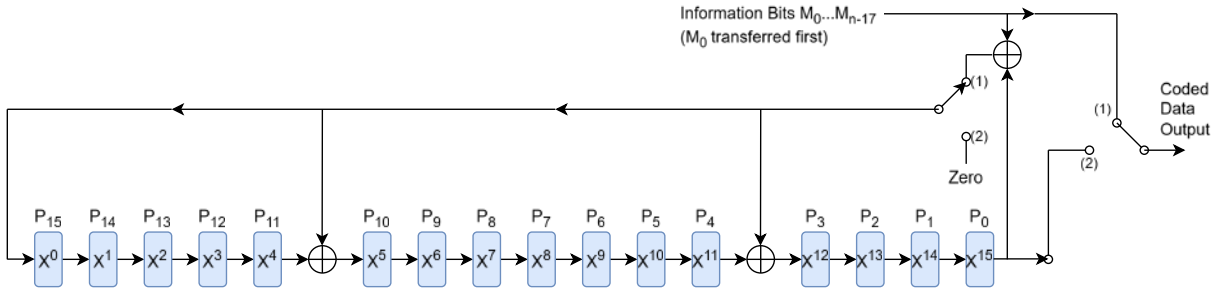


Figure 7-2: Logic diagram of CRC-16 generator

The CRC-16 adds error identification and increases the overall accuracy of the system. Comparing the captured CRC-16 to the calculated CRC-16 of the captured data on the ground allows for errors in the data stream to be identified. Identified erroneous frames can be retransmitted to ensure accurate captured data.

7.2 ENCODING & MODULATION

TM Transfer Frames can be be encoded and modulated using either of the following encoding & modulation schemes:

- CCSDS
- DVB-S2

A brief overview of these are given in the subsections below.

7.2.1 CCSDS

CCSDS encoding and modulation has been implemented in line with the CCSDS RECOMMENDED STANDARD FOR TM SYNCHRONIZATION AND CHANNEL CODING [3].

Figure 7-3 below shows a brief overview of the CCSDS encoding & modulation path.

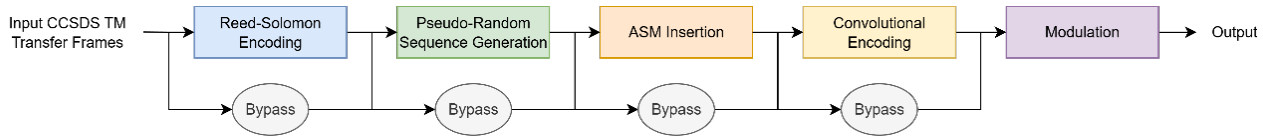


Figure 7-3: CCSDS encoding & modulation path.

7.2.1.1 REED-SOLOMON ENCODER

The RS encoder is used to correct burst errors in the communication link. For maximum performace (at the expense of accompanying overhead, the E=16 option can correct 16 R-S symbols in error per codeword. For lower overhead (with reduced performacne), the E=8 option can correct 8 R-S symbols per codeword.

The implemented CCSDS managed parameters is listed in Table 7-5 below.

Table 7-5: CCSDS Reed Solomon encoder managed parameters.

Managed Parameter	Value
Error Correction Capability (E, symbols)	8, 16
Interleaving Depth (I)	1, 2, 3, 4, 5, 8
Virtual Fill Length (Q, symbols)	0

7.2.1.2 SCRAMBLER OR PSEUDO RANDOMIZER

The scrambler is used to ensure there is sufficient bit transitions in the encoded data stream for the receiver to be able to lock. The CCSDS polynomial used is:

$$h(x) = x^8 + x^7 + x^5 + x^3 + 1$$

Figure 4 is the illustrated example from [3] to show how the polynomial can be implemented logic wise.

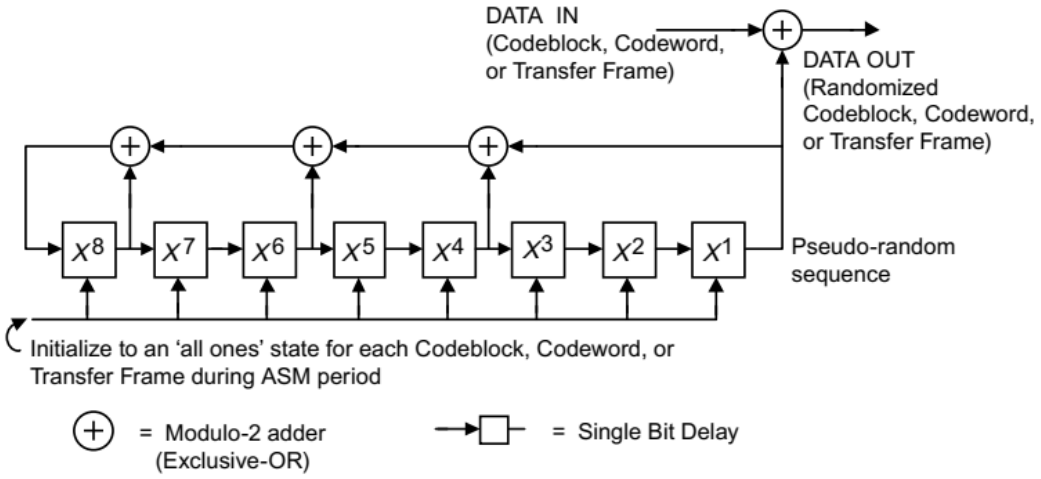


Figure 7-4: CCSDS Pseudo randomizer.

This logic block is reset at the start of a transfer frame.

7.2.1.3 ASM INSERTER

The attached sync marker is a 32-bit marker that is attached to the start of each transfer frame. The CCSDS marker used is 0x1ACFFC1D as defined in [3]. This marker is used to synchronize the scramblers and RS decoders on the receive side.

Table 7-6: CCSDS ASM marker managed parameters.

Managed Parameter	Value
ASM Length (bits)	32

7.2.1.4 CONVOLUTIONAL ENCODER

The convolutional encoder is used for forward error correction of white Gaussian noise. This encoder is a half rate encoder which means the data rate at the input is half the data rate of the output. The connection vectors G1 = 1111001 (171 octal) and G2= 1011011 (133 octal) are used.

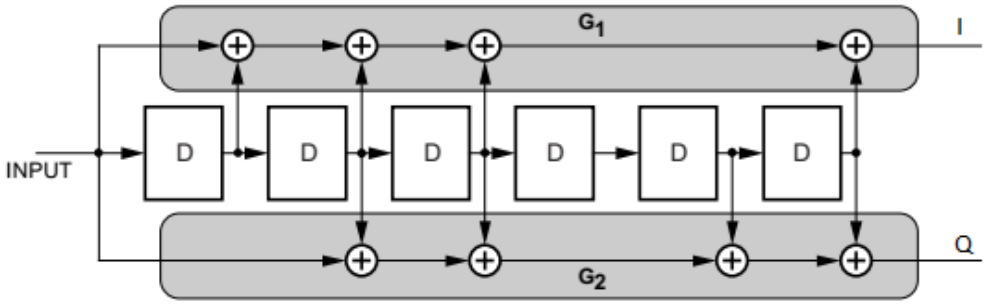


Figure 7-5: Convolutional encoder shift register.

The Q output arm is not inverted as is sometimes the case for non-punctured encoders.

The implemented CCSDS managed parameters are listed in Table 7-7 below.

Table 7-7: CCSDS Convolutional encoder managed parameters.

Managed Parameter	Value
Code rate	1/2, 2/3, 3/4, 5/6, 7/8

7.2.2 DVB-S2

DVB-S2 encoding and modulation has been implemented in line with the ETSI EN 302 307-1 V1.4.1 (2014-11) [5] standard.

Figure 7-6 below shows a brief overview of the DVB-S2 encoding & modulation path.



Figure 7-6: DVB-S2 encoding & modulation path.

The DVB-S2 encoding is implemented in firmware, with the following permanent managed parameters that are kept static for a given mission:

Table 7-8: DVB-S2 mission static managed parameters.

Static Managed Parameter	Supported Values
Transmission mode	VCM
Baseband pulse shaping roll-off factor	0.2, 0.25, or 0.35
Dummy PLFRAME utilization	No, a continuous data stream is generated with OID transfer frames when no data is available.
Scrambling code number n	Integer: 0
Number of MODCOD mission phase	Integer: 28
Supported FECFRAME size	Short, Normal
Supported pilot insertion status	ON, OFF

The variable parameters, shown in Table 7-9, are imbedded in the PL-Header of the transmitted signal, and consequently it is not needed to provide them to the receiver when working in VCM mode.

Table 7-9: DVB-S2 variable managed parameters.

Variable Managed Parameter	Supported Values
Current MODCOD	Integer: 1 to 28
Current FECFRAME size	Short or Normal
Current pilot insertion status	ON or OFF

Commercial ground station networks that support DVB-S2 usually output the stream at the BB-frame level which means the upstream CCSDS packetization needs to be removed.

A 4-byte ASM marker (0x1ACFFC1D) provides a sync marker before each transfer frame header to align the packet header to extract the payload data.

The data breakdown between the different levels of packet structures when using DVB-S2 encoding is given in Figure 7-7 below. The lowest level being the raw payload data input from the data source.

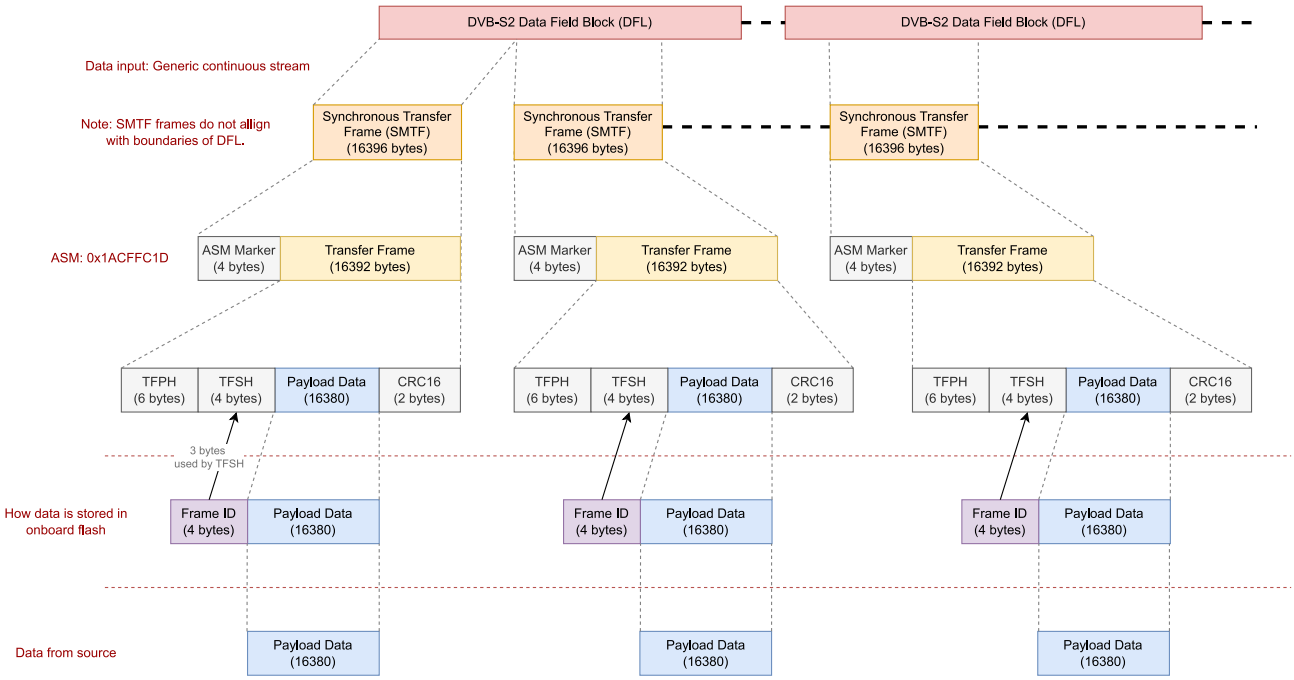


Figure 7-7: DVB-S2 packet structure with CCSDS transfer frames.

8 TRANSMITTER FEATURES

The VestaLink S25 has several features that allow for versatile transmitter operation. The available features are broken down into the following subsections:

- 8.1 Transmitter Communication
- 8.2 Configuration and Control
- 8.3 Payload Data Routing
- 8.4 Memory Interface
- 8.5 Data Debugging Tools
- 8.6 Telemetry and Monitoring
- 8.7 System Protection Limits

8.1 TRANSMITTER COMMUNICATION

The VestaLink S25 has an I2C, CAN and UART interface that can be used (if selected) to operate the payload. The communication interfaces also have status registers (Sections 9.2.2.1, 9.2.2.3, and 9.2.2.7) that can be polled to determine if any of the previous communication interactions failed. The status information consists of 5 flags:

- **Invalid Address:** This flag indicates a previous message tried to read from or write to an invalid address.
- **Out of Range:** This flag indicates a previous message was an incorrect size.
- **Not Writeable:** This flag indicates a previous message tried to write to a read-only register address.
- **Internal Error:** This flag indicates that an internal error occurred during the processing of a previous message.
- **Buffer Overflow:** This flag indicates that the queue handling the messages is full and a message has been dropped.

8.2 CONFIGURATION AND CONTROL

The VestaLink S25 can be configured and controlled with the registers in the “Control” and “PA Status and Control” sections of the register map (Section 9). These registers control the radio operating modes, the RF Channel and the Power Control loop.

8.2.1 OPERATING MODES

The VestaLink S25 has four operating modes that it can be set to. The operating modes and the order of mode changes are shown in Figure 8-2. The order of mode changes will always be followed e.g. if the board is in Standby and a command is received to go to Transmit, the radio will first step to Config mode and then to Transmit mode. The order of mode changes ensures that the radio is properly initialized or shut down during operation.

The four operating modes are:

- **Standby Mode**: The VestaLink S25 starts up in this mode (the radio’s lowest power mode). During this mode only the FPGA is active, while all the other peripherals are disabled. Board telemetry is active in Standby mode, however certain sections’ power/temperature readings will display all zeroes while its section is not powered.
- **Flash Access:** This mode allows for the on-board flash memory to be accessed while the rest of the peripherals are powered down. The high-speed payload data interface and memory interfaces are enabled during Flash Access mode. Payload data can be stored in/read out of memory through the data interface in a low power state.
- **Config mode**: All the peripherals and RF subsystems, except for the PA, are enabled and initialized. During this mode the configuration values can be changed.
- **Transmit mode**:The PA is turned on, the DAC is enabled and the RF signal is transmitted.

Certain features of the transmitter are only accessible/active in specific operating modes. Figure 8-1 shows what features are accessible in each transmitter operation mode.

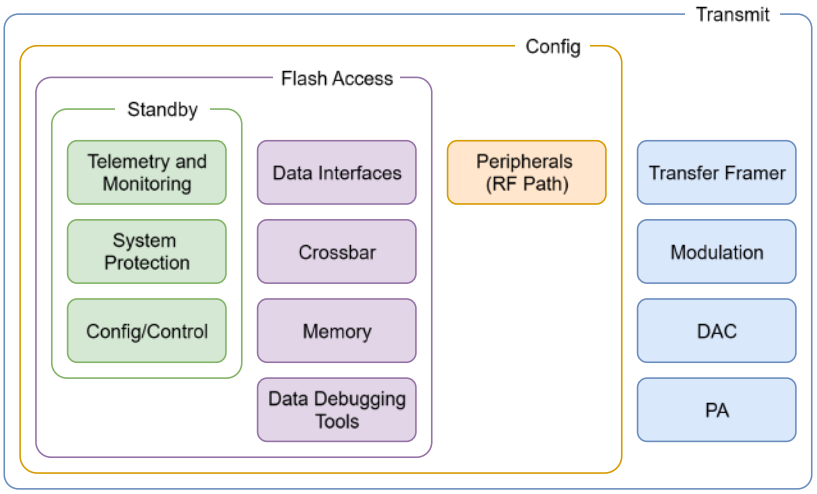


Figure 8-1: VestaLink S25 feature accessibility breakdown.

The transmitter must be instructed to go back to Standby mode before it is powered off to ensure a proper shutdown procedure is followed. Going into Standby mode properly powers down the on-board flash memory and the other peripherals on the transmitter.

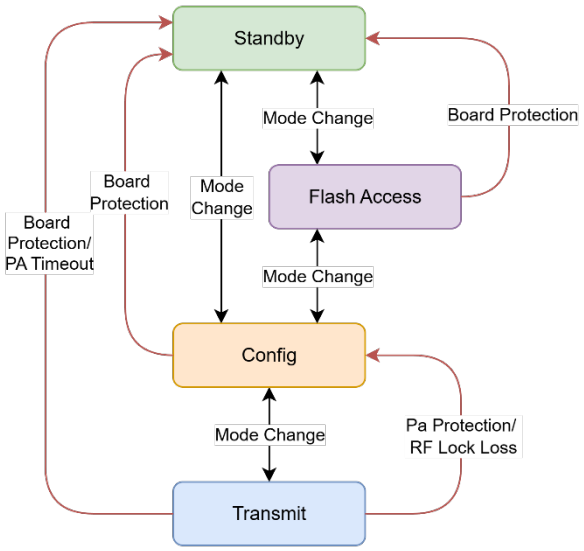


Figure 8-2: VestaLink S25 mode change overview.

The VestaLink S25 has protection limits that will automatically step the mode to a safer one if the protection limit is triggered. The protection limits are discussed in Section 8.7.

8.2.2 RF CHANNEL

The RF channel can be configured for a variety of operating conditions. These configuration settings are stored in the following registers:

- **Symbol Rate** Register (Section 9.2.5.2): sets the symbol rate at which the VestaLink S25 will transmit. The available symbol rate range for the VestaLink S25 is 100 - 5000 ksps in 100 ksps steps.
- **Frequency** Register (Section 9.2.5.3): sets the frequency at which the VestaLink S25 will transmit. The available frequency range for the VestaLink S25 is 2150 MHz – 2290 MHz in 1 MHz steps.
- **Output Power** Register (Section 9.2.9.2): sets the output power at which the VestaLink S25 will transmit. The available output power range for the VestaLink S25 is 27 – 33 dBm in 1 dBm steps.
- **Encoding Control** Register (Section 9.2.5.4): sets the encoding & modulation scheme to use as well as the parameters common to both CCSDS and DVB-S2 encoding & modulation schemes.
- **DVB-S2 Config** Register (Section 9.2.5.5): sets the DVB-S2 encoding parameters. The DVB-S2 core adds the BB-frame header to the data stream according to the CCSDS standard.
- **CCSDS Config** Register (Section 9.2.5.6): sets the CCSDS encoding parameters.

The above registers should be configured when the transmitter is in the Config state, except for the encoding register that can also be dynamically changed while transmitting.

8.2.3 POWER CONTROL LOOP

The VestaLink S25 has a PID controller that maintains a constant average RF output power when transmitting. The PA controller accomplishes this by measuring the RF output power and PA temperature and feeding back the error signal to the PA. The hardware section of the power control loop is shown Figure 8-3. Each VestaLink S25 is tested and calibrated to provide a constant output power across the operating temperature range.

The transmitter also supports several open- and closed loop control modes (Table 8-1). The specific mode of the control loop is set in the PID mode field of the PA_CONF register (Section 9.2.9.2).

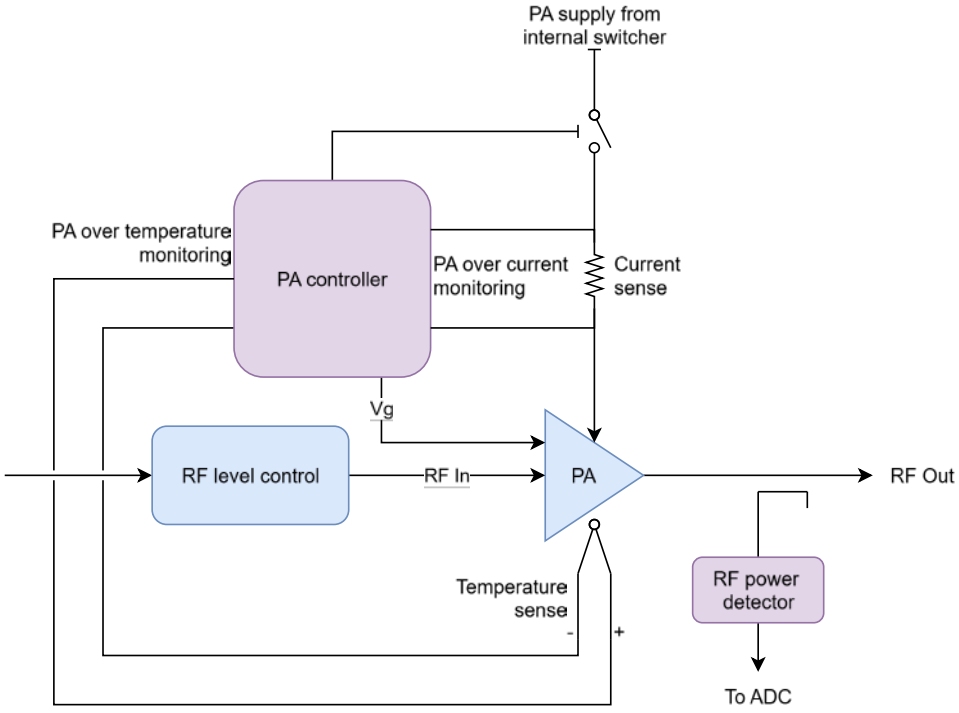


Figure 8-3: PA control loop overview.

The PA controller implements the power up and power down sequencing of the PA and provides a hardware level implementation (processor independent) for PA over current and over temperature protection. The RF output power is set in the PA_CONF register (See Section 9.2.9.2).

Table 8-1: PA control modes

PID Mode	PA0_Conf1 Register Value
Off	0x00
Power Detector with Temperature	0x01
Power Detector only	0x02
Open Loop with Temperature	0x03
Open loop	0x04

8.3 PAYLOAD DATA ROUTING

The VestaLink S25 provides a highly configurable data routing network to cater for a variety of transmitter operating conditions. The payload routing network is shown in Figure 8-4.

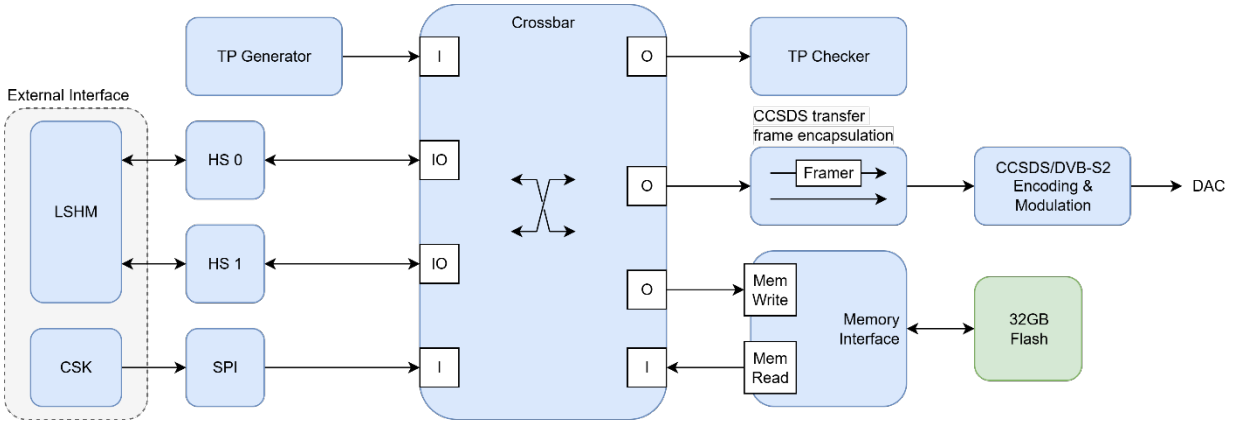


Figure 8-4: Payload data routing for the VestaLink S25 .

The payload data routing network can be broken down into 5 subsections:

- 8.3.1 Payload Data Interface
- 8.3.2 Data Debugging Tools
- 8.3.2 Routing Crossbar
- 8.4 Memory Interface
- 8.5.3 Transfer Frame Encapsulator

8.3.1 PAYLOAD DATA INTERFACE

The VestaLink S25 receives the payload data to be transmitted via the data interface. See document “ICD-01-10022” for more details on the physical implementation and protocol descriptions of the available external data interfaces. The data interface consists of two/three physical connectors: the PC104 connector and the LSHM high speed connector.

The PC104 connector has a simple low speed SPI interface to be used as a data receive interface. The SPI interface is fixed and available if chosen in the option sheet (OPT-01-10022).

The LSHM high speed connector provides a physical connection to two high speed data interfaces. The high speed (HS) interfaces will be referred to as nodes. The nodes’ data transfer protocols are fixed during operation and must be chosen beforehand. The data transfer protocols implemented are chosen using the option sheet (OPT-01-10022).

The data interface configuration and status registers are available in Section 9.2.3. These registers can be used to configure and monitor the two HS nodes and the SPI interface.

All data interfaces are data receivers, but some have data transmission capabilities. SpaceWire (Figure 8-5) can send and receive data while CUBECOM LVDS (Figure 8-6) and SPI can only receive data.

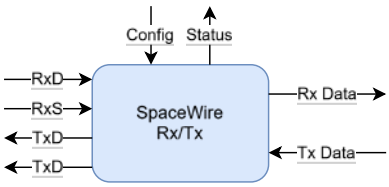


Figure 8-5: Example of SpaceWire RX/TX data interface.

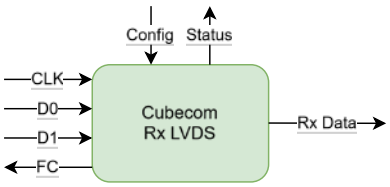


Figure 8-6: Example of LVDS RX-only data interface.

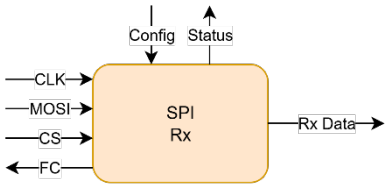


Figure 8-7: Example SPI RX-only interface.

An example of the data interfaces and crossbar being used to mux the outputs of 2 data sources is shown in Figure 8-8. The two sources can be connected to the transmitter via the LVDS interfaces and the active source that is transmitted can be selected using the routing crossbar.

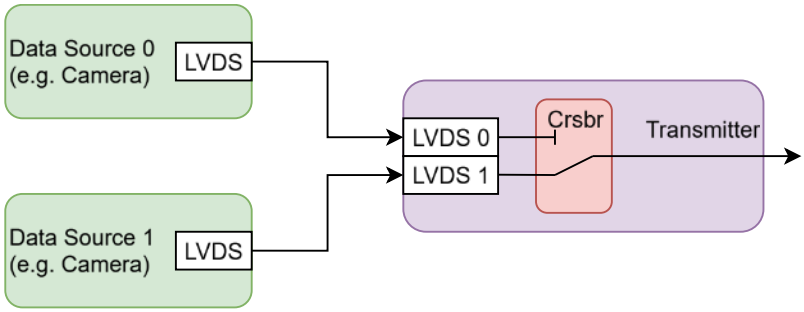


Figure 8-8: Example implementation using dual LVDS interface.

Protocols with transmission capabilities can be used to route data out of the VestaLink S25 using the routing crossbar. Figure 8-9 shows an example of a SpaceWire implementation that connects a data source to an OBC (or other data processing module) using the data interfaces. The data interfaces are connected to each other routing the data through the transmitter.

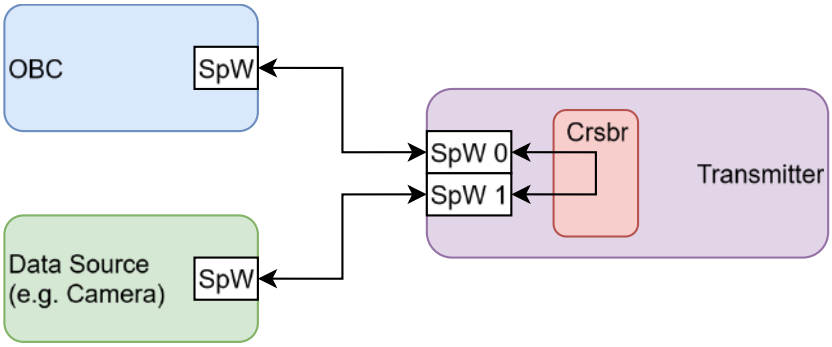


Figure 8-9: Example implementation using dual SpaceWire interface.

8.3.2 ROUTING CROSSBAR

The routing crossbar controls where the data flows inside the transmitter. The crossbar allows for a single data input to be connected to one or more data outputs (In case of multiple outputs connected to a single input the throughput rate regarding the data outputs will be determined by the slowest output). The control of the routing crossbar is further discussed in Section 9.2.8.7.

A standard use case is shown in Figure 8-10 where the input of HS0 is routed to the transfer frame encapsulator to be transmitted out of the VestaLink S25. Multiple outputs can be connected to a single input and different parallel routes can also be linked simultaneously (Figure 8-11 shows a more complex routing configuration).

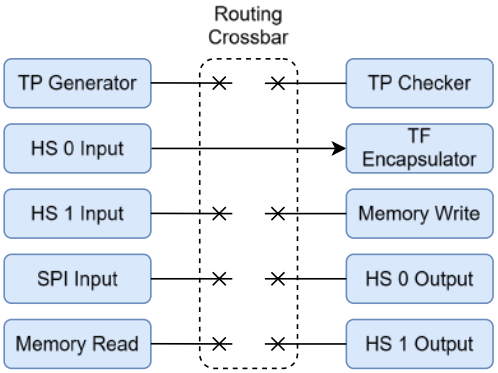


Figure 8-10: Simple single route.

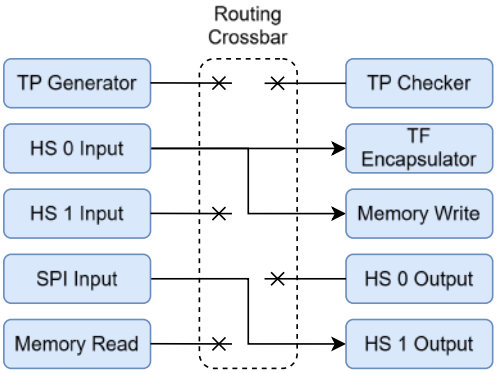


Figure 8-11: Example of complex parallel routing.

Note only one memory interface operation can run at a time. Memory read and write operations cannot run in parallel.

8.4 MEMORY INTERFACE

The on-board flash memory uses 32GB eMMC flash to allow for payload data to be transferred to the internal memory before a downlink session. The stored data can then be read from the flash memory at the maximum transmission speed of the VestaLink S25 . This has the benefit of allowing a lower speed data interface to be used while still being able to transmit at the maximum transmission rate.

The VestaLink S25 can transmit up to 16 GB in a typical 10-minute LEO pass at the maximum data rate. Utilizing the on-board memory allows for the optimal use of a downlink pass regardless of data interface speed. Using the internal memory also allows for easy retransmission of any individual packets that were corrupted or lost during a downlink.

The eMMC flash has automatic bad block detection and will replace bad blocks with spare replacement blocks. The on-board memory health information can be found in registers detailed in Section 9.2.7.3.

The on-board memory is broken down into 16 KB data segments that consist of the payload data and the packet ID. These packets are addressed using memory indexes. The packet ID is the memory index and is used to trace the packet back to where in memory it was read from. The breakdown of the memory structure is shown in Figure 8-12.

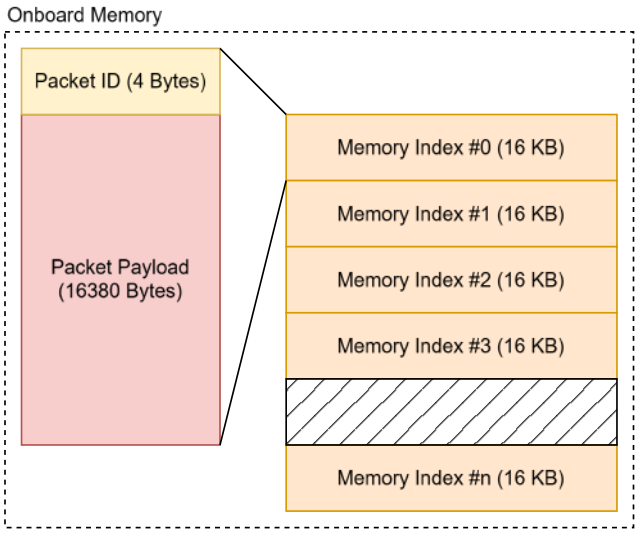


Figure 8-12: On-board flash memory packet and index breakdown.

All control and status registers related to the memory interface can be found in Section 9.2.7.

8.4.1 MEMORY WRITE

The VestaLink S25 stores the payload data in 16384-byte packets. The first 4 bytes of each data packet are used to store a packet ID number which is the flash memory block index followed by 16380 bytes of payload data. The packet ID is auto incremented during writing. The packets are incrementally stored in memory as shown in Figure 8-13. The packet ID is used to match the data packet with where in the memory that packet is stored.

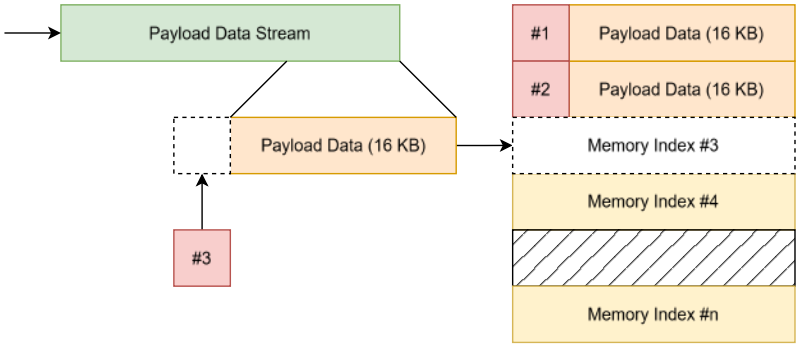


Figure 8-13: On-board flash memory write operation.

The EOF is added to the final packet written into memory when the write successfully completes from a manual or automatic stop condition. If a stop is triggered before buffer is full, the interface will pad the rest of the packet with "0xAA" bytes to fill it to 16 KB. As described in Section 7.1.2, the 24th bit of the packet ID is the EOF flag, followed by 23 bits for the packet index number. The EOF flag will cause the data encapsulation block to flush any data out and embed the flag in the Transfer Frame Secondary Header which serves as a signal to the receiver when an EOF occurred.

Registers related to the memory write interface can be found in Sections 9.2.7.4 to 9.2.7.10.

8.4.2 MEMORY READ

After the payload data is stored in the flash memory, it is ready to be read out and transmitted. The stored payload can be read out at the maximum transmission rate of the VestaLink S25. The memory read operation is shown in Figure 8-14. The read operation increments through the memory indexes, starting at the chosen index, reading each packet out after the other.

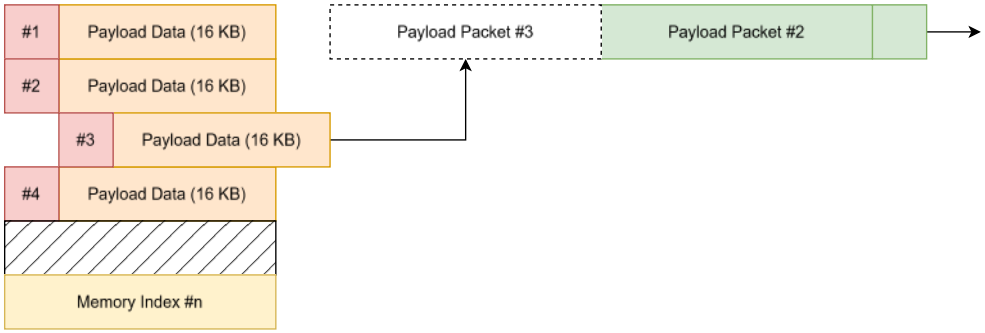


Figure 8-14: On-board flash memory read operation.

The Packet ID is also read with the raw payload data and stored in the Transfer Frame Secondary Header. The Packet ID can be used on the ground to determine data validity (By checking if there are any missing IDs in the data stream). The missing IDs can be requested directly by re-transmitting the exact missing ID as it represents the exact memory location of the data packet.

Registers related to the memory read interface can be found in Sections 9.2.7.11 - 9.2.7.16.

8.4.3 MEMORY RETRANSMIT

The memory interface has retransmission capability where the OBC can request retransmission of any required packets during or after an ongoing memory read process. Figure 8-15 shows an example of a data transmission operation where a missing packet is detected, a retransmission is requested, and the missing packet is sent during the current read operation.

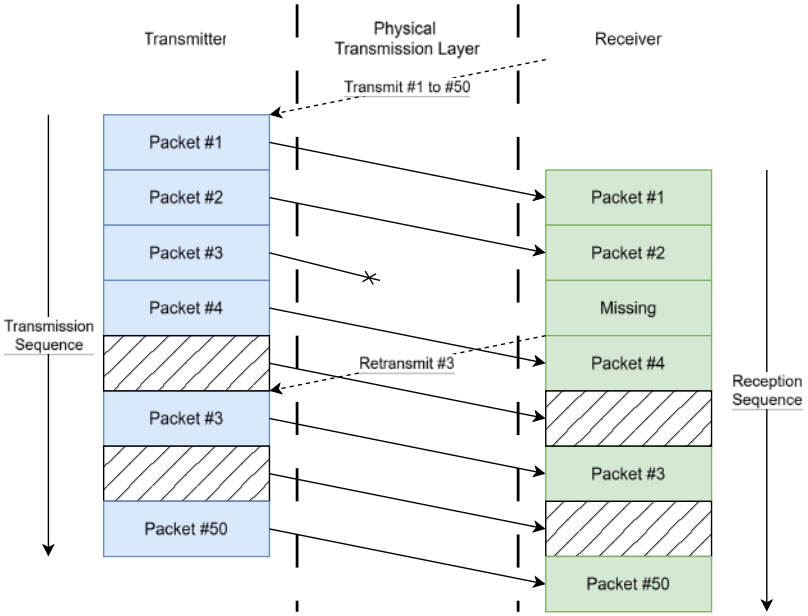


Figure 8-15: Example use case of the retransmission feature

Retransmission packets can be requested by writing to the retransmit queue register using the packet ID (3 bytes) and the number of consecutive packets (1 byte). The requests are inserted into a queue (FIFO) of up to 255 requests. The queued packets can be dynamically injected into the active memory read data stream (If set to a higher priority), or it can be sent after the data transmission is complete.

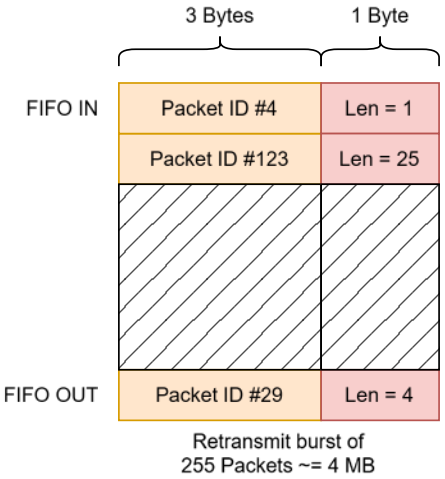


Figure 8-16: Retransmit structure

Registers related to the memory retransmit interface can be found in Sections 9.2.7.17 - 9.2.7.21.

8.5 DATA DEBUGGING TOOLS

The test pattern generator and test pattern checker can be used to debug and test the payload data routing network. A simple debugging route is shown in Figure 8-17.

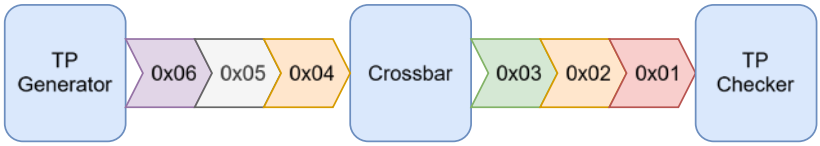


Figure 8-17: Simple data route debugging configuration.

8.5.1 TEST PATTERN GENERATOR

The function of the test pattern generator is to create an easily verifiable stream of data. Using the routing crossbar the test pattern can be written into memory, sent to the transfer frame encapsulator to be transmitted, written into the test pattern checker, or connected to and written out of a transmit capable HS node. These various routing paths can be used to test and debug the different write/transmit functions of the payload routing network.

The test pattern generator has two configurations: It can implement a PRBS7 generator or an 8-bit counter. The PRBS7 generator, shown in Figure 8-18, is a standard linear feedback shift register (LFSR) with an 8-bit polynomial ($x^7 + x^6 + 1$). The polynomial is seeded with binary value 0x02 by default.

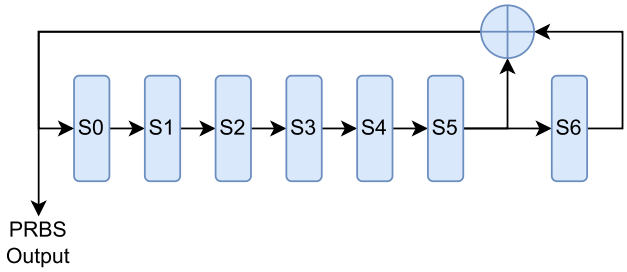


Figure 8-18: PRBS7 generator shift register.

See Sections 9.2.8.8 - 9.2.8.10 for more information on how to use the test pattern generator.

8.5.2 TEST PATTERN CHECKER

The test pattern checker performs a complementary function to the test pattern generator. The test pattern checker can be used to verify simple verifiable streams of data. The test pattern checker can be set up to check for a PRBS7 (same as one generated using Figure 8-18) or a simple 8-bit counter.

The test pattern checker can be fed a data stream from any available input connected to the routing crossbar. These various routing paths can be used to test and debug the different read/receive functions of the payload routing network.

See Sections 9.2.8.11- 9.2.8.16 for more details on how to use the test pattern checker.

8.5.3 TRANSFER FRAME ENCAPSULATOR

The payload data framing discussed in the data encapsulation process (Section 7.1) starts with the transfer frame encapsulator. The transfer frame encapsulator accepts incoming payload data, encapsulates it in the transfer frame structure and writes it to the DVB-S2 modulator that then encodes the data.

The DVB-S2 buffer will transmit dummy frames if it does not have enough data in its buffer to create a data field block. To prevent dummy frames from being generated the transfer frame encapsulator always keeps the DVB-S2 data buffer full. The transfer framer accomplishes this by creating OID data packets if there isn't enough payload data in the framer's own data buffer (Basic flow of data shown in Figure 8-19).

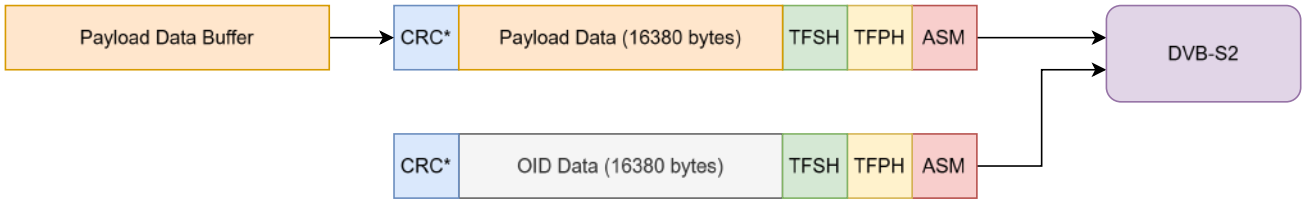


Figure 8-19: Basic flow of the transfer frame encapsulator.

8.5.3.1 DATA BUFFER FLUSHING

The VestaLink S25 has two main modes to send user data to the transfer frame encapsulator: either from flash memory or from the high-speed data interface. Depending on the mode, some data could be stuck in the transfer frame encapsulator buffer. To prevent misalignment of memory read packets or to just remove random data in the buffer, the buffer can be flushed. This can be achieved by setting the Flush_Data bit the Transfer Frame Control register (See Section 9.2.8.1)

- From Flash Memory – The data is read as blocks which align exactly with the packet lengths, so no flushing is required afterwards. Flushing before a memory read is started if the transfer frame buffer is not empty is required to align the memory packets with the framer to prevent misalignment issues.
- From Data source – Depending on the data source and the control characters supported, it might be needed to flush the VestaLink S25.

Any EOF received from the input data stream will trigger the buffer to flush and add the EOF to the Transfer Frame Secondary Header. Flushing the data buffer pads the rest of the packet length with "0xAA" bytes to create a full transfer frame packet to be read out. Reading out a full data packet empties the data buffer.

The transfer frame encapsulator buffer count can be read after a data transmission to determine if a flush is required. The transfer frame buffer count can be found in the TF_Buf_Cnt register.

8.6 TELEMETRY AND MONITORING

The available telemetry of the VestaLink S25 consists of the board status, radio temperature and subsystem currents/voltages. The available telemetry is discussed in further detail below.

8.6.1 BOARD STATUS

The board status register (Section 9.2.4.1) gives an overview of the operational status and error conditions for the VestaLink S25 and is the only register that needs to be actively monitored.

The status flags of the board status register:

- **I2C_Status**: This flag is set if any faults are detected regarding the I2C communication link (Section 9.2.2.1).
- **CAN_Status**: This flag is set if any faults are detected regarding the CAN communication link (Section 9.2.2.3).
- **UART_Status**: This flag is set if any faults are detected regarding the UART communication link (Section 9.2.2.7).
- **MEM_Status**: This flag is set if the memory fails to initialize. (Section 9.2.7.1).
- **RF_Status**: This flag is set if the RF subsystem loses lock. (Section 9.2.4.2).
- **PA_Status**: This flag is set if any faults are detected regarding the PA. (Section 9.2.9.1).
- **PA_Warn_Status**: This flag is set when the PA temp exceeds the warn temp limit (Section 9.2.9.3).
- **Timeout_Status**: This flag is set when the transmitter has been transmitting for more than the set time limit (Section 9.2.9.4).
- **Brd_Warn_Status**: This flag is set when the board temperature exceeds the warn temperature (Section 9.2.6.4).
- **Brd_Trip_Status**: This flag is set when the board temperature exceeds the trip temperature limit (Section 9.2.6.4).
- **Brd_Curr_Status**: This flag is set when a current sensor is reporting an anomalous reading (Section 9.2.4.4).
- **Brd_Volt_Status**: This flag is set when a voltage sensor is reporting an anomalous reading (Section 9.2.4.5).

8.6.2 TEMPERATURE MONITORING

The VestaLink S25 has multiple temperature sensors to provide a comprehensive temperature monitoring system. The temperature values can be read from the temperature registers (Section 9.2.4.3). Table 8-2 provides a list of the temperature sensors and their respective locations on the VestaLink S25.

Table 8-2: Temperature Monitoring Points

Temp Sensor	Description	Location
0	PA	Top case of the PA.
1	PA SMPS	Adjacent to PA DC-DC converter.
2	PA Controller (Board 0)	Power controller's IC's internal temperature sensor.
3	FPGA	FPGA's internal temperature sensor.
4	Board 1	Underside of VestaLink S25.
5	Board 2	Underside of VestaLink S25.
6	Board 3	Underside of VestaLink S25.

8.6.3 POWER DISTRIBUTION

The VestaLink S25 has an extensive diagnostic measurement capability which enables voltage and current monitoring for every DC/DC switcher and regulator on the transmitter. The currents are monitored by an independent soft processor running on the FPGA. The power distribution and monitoring points are shown in Figure 8-20.

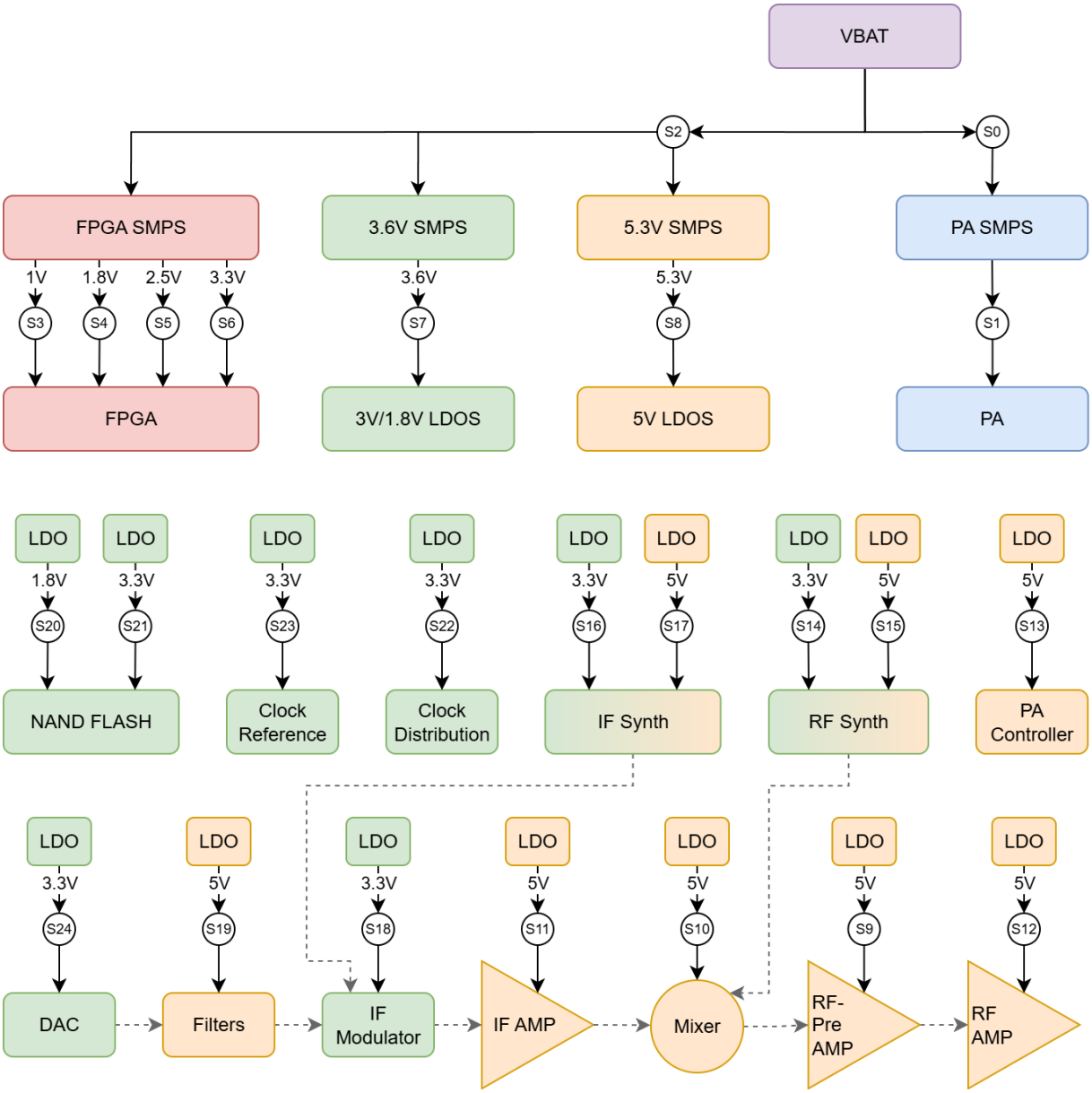


Figure 8-20: Power network with monitoring points.

The sensor monitoring points shown in Figure 8-20 are listed in Table 8-3 below.

Table 8-3: Power network monitoring points.

Sensor	Description	Voltage
0	PA SMPS Input	7 V - 20 V
1	Regulated 21 V OF PA SMPS	~21 V
2	VBAT board input	7 V - 20 V
3	Regulated 1 V LDO for FPGA	1V
4	Regulated 1.8 V LDO for FPGA	1.8 V
5	Regulated 2.5 V LDO for FPGA	2.5 V
6	Regulated 3.3 V LDO for FPGA	3.3 V
7	Regulated 3.7 V OF SMPS	3.7 V
8	Regulated 5.3 V OF SMPS	5.3 V
9	-	-
10	-	-
11	-	-
12	Regulated 5 V LDO for RF driver amplifier	5 V
13	Regulated 3.3 V LDO for IF synthesizer	3.3 V
14	-	-
15	-	-
16	Regulated 5 V LDO for IF synthesizer charge pump	5 V
17	Regulated 5 V LDO for PA controller	5 V
18	Regulated 3.3 V LDO for modulator	3.3 V
19	Regulated 5 V LDO for baseband filter	5 V
20	Regulated 1.8 V LDO for flash memory	1.8 V
21	Regulated 3.3 V LDO for flash memory	3.3 V
22	Regulated 3.3 V LDO for clock distribution	3.3 V
23	Regulated 3.3 V LDO for clock reference	3.3 V
24	Regulated 3.3 V LDO for digital to analog converter	3.3 V

8.7 SYSTEM PROTECTION LIMITS

The VestaLink S25 has system protection limits implemented to automatically safeguard it from extreme operating conditions (Operating outside of recommended operating conditions).

The system protection limits are:

- 8.7.1 Temperature limits (PA and board).
- 8.7.2 PA current limit.
- 8.7.3 PA transmit time limit (PA Timeout).

8.7.1 TEMPERATURE LIMITS

The temperature of the board and PA are monitored by the VestaLink S25 and compared against set limits. There are 3 temperature setpoints (limits) for the PA temperature. The first two levels can set warning flags and automatically limit the output power to a predefined power level (If the set power level is 0 then the board will return to Config mode if triggered). The third level will switch the RF off and return the VestaLink S25 to Config mode.

The board temp limit triggers the VestaLink S25 to return to Standby mode and sets a warning flag when the measured board temperature exceeds the set limit. All the temperature protection limits can by bypassed by setting the Disable_Overtemp bit to '1' or each limit can be individually bypassed by setting the limit to the value of 255. By bypassing a protection limit the board will no longer respond to that specific extreme condition and the transmitter could be made to operate in conditions outside the defined temperature limits, potentially damaging it. The temperature protection levels are configured using the registers detailed in Section 9.2.6. A summary of the temperature limits of the VestaLink S25 is given in Table 8-4.

Table 8-4: Temperature Limits Overview.

Temperature Limit	Default Trip Level [°C]	Default Action
PA Limit 1	95	Limits output RF power to 30 dBm
PA Limit 2	97	Limits output RF power to 27 dBm
PA Limit 3	100	Stop RF transmission and return to Config mode.
Board Limit	75	Returns to Standby Mode

8.7.2 PA CURRENT LIMIT

The PA current limit protects the PA from operating at damaging currents for too long. When the measured current exceeds the set limit, the PA will be switched off, the radio will be returned to Config mode, and the status flag will be updated. The over current trip is implemented on a hardware level.

8.7.3 PA TIMEOUT

There is also a configurable transmit timeout which will return the transmitter back to Standby mode after 18 minutes. The timeout protects the radio against staying excessively long in Transmit mode if communication to the radio is lost during data transmission. The timeout can be bypassed to allow for indefinite transmission times (not recommended). See the timeout register in Section 9.2.9.4. A flag is set in the status register when the transmitter timeouts (see section 9.2.4).

9 REGISTER MAP

All user-accessible configurations are stored in the register map. All registers are 32-bit wide (4 bytes) and are addressed using a 16-bit address. Some registers are read-only and will ignore write commands. There are a few registers which are usually read-only but can be written to under the correct conditions. Register addresses not listed in the register map are reserved for future use or internal CUBECOM use-cases. Therefore, care should be taken not to read/modify any registers not listed here, accept if instructed to do so by a qualified CUBECOM engineer and/or technical representative..

Section 9.1 gives an overview of all the registers. Section 9.2 goes into further detail regarding the available registers.

9.1 REGISTER OVERVIEW

Table 9-1 below is a summary of all registers.

Table 9-1: Register map overview

GENERAL REGISTERS

ADDRESS	Name	Description	Reset
0X0001	*ID*	Identifier for the VestaLink S25.	0x53545832
0X0002	*SERIAL*	Serial number for this board.	-
0X0003	*SW_Version*	Software version; Major, Minor and Patch.	-
0X0004	*FW_Version*	Firmware version; Major, Minor and Patch.	-
0X0005	*HW_Version*	Hardware version; Major, Minor and Patch.	-
0X0006	*UPTIME*	This value increments continuously, can be used to check comms.	0x00000000
0X0007	*SCRATCHPAD*	Test register for initial comms read/write debugging.	0x00000000

COMMUNICATION REGISTERS

ADDRESS	Name	Description	Reset
0X0010	*I2C_STAT*	I2C Status register. Flags showing Busy / Invalid / Out of Range/ Overflow. Cleared on read.	0x00000000
0X0011	*I2C_CONF*	I2C communication link configuration	-
0X0012	CAN_STAT	CAN Status registers. Flags showing Busy / Invalid / Out of Range/ Overflow. Cleared on read.	0x00000000
0X0013	*CAN_CONF*	CAN0 communication link configuration.	-
0X0326/ 0X0327	CAN_CSP_CONF	Cubesat Space Protocol configuration	-
0X0328	CANTS_CONF	CAN-TS configuration.	-
0X0014	*UART_STAT*	UART Status register. Flags showing Busy / Invalid / Out of Range/ Overflow. Cleared on read.	0x00000000
0X0015	*UART_CONF*	UART communication link configuration	-
0X001F	*COMM_UNLOCK*	Communication link configuration access control.	0x00000000

DATA SOURCE REGISTERS

ADDRESS	Name	Description	Reset
0X0020/ 0X0026	*HS_NODE_INFO*	High speed data interface information	0x00000000
0X0021/ 0X0027	*HS_CTRL*	Controls the high-speed data interface.	0x00000000
0X0022/ 0X0028	*HS_STAT_0*	Displays status information of the high-speed data interface	0x00000000
0X0023/ 0X0029	*HS_STAT_1*	Displays extra status information of the high-speed data interface	0x00000000
0X0024/ 0X002A	*HS_RX_CNT*	Displays received byte count of the high-speed data interface	0x00000000
0X0025/ 0X002B	*HS_TX_CNT*	Displays transmitted byte count of the high-speed data interface	0x00000000
0X002C	*SPI_CTRL*	Controls the SPI interface.	0x00000000
0X002D	*SPI_STAT*	SPI link and usage status.	0x00000000
0X002E	*SPI_RX_CNT*	Displays count of the number of 32-bit words received and passed through the SPI data interface.	0x00000000

TELEMETRY REGISTERS

ADDRESS	Name	Description	Reset
0X0030	*BOARD_STAT*	General status of the device.	0x00000000
0X0031	*RF_STAT*	IF Synth Lock, RF Synth Lock.	0x00000000
0X0032-0X0035	*TEMP*	Board, FPGA, PA and PA SMPS temperature measurements.	0x00000000
0X003E	*CURRENT_ERR*	Bit flags for current sensor out of range errors.	0x00000000
0X003F	*VOLTAGE_ERR*	Bit flags for voltage sensor out of range errors.	0x00000000
0X0040-0X004F	*CURRENTS*	Range of registers containing monitored current values.	0x00000000
0X0050-0X005F	*VOLTAGES*	Range of registers containing monitored voltage values.	0x00000000

CONTROL REGISTERS

ADDRESS	Name	Description	Reset
0X0060	*MODE*	Controls the state of the transmitter. Standby / Flash Access / Config / Transmit.	0x00000000
0X0061	*SYMBOL_RATE*	Symbol Rate. Range: 100 - 5000 ksps in 100 ksps steps.	-
0X0062	*FREQUENCY*	Frequency. Range: 2150 MHz – 2290 MHz in 1 MHz steps	-
0X0065	ENCODING_CTRL	Controls the universal encoding settings.	-
0X0066	DVBS2_CON	Controls the DVB-S2 specific settings.	-
0X0067	CCSDS_CONF	Controls the CCSDS specific settings	-
0X0068	ENCODING_STAT	Contains count of dummy frames generated by DVB-S2 encoder when no data is available.	0x00000000

THERMAL PROTECTION REGISTERS

ADDRESS	Name	Description	Reset
0X0070	*PROTECTION_LEVEL_1*	PA over temperature protection level 1.	0x001E005F
0X0071	*PROTECTION_LEVEL_2*	PA over temperature protection level 2.	0x001B0061
0X0072	*PROTECTION_LEVEL_3*	PA over temperature protection level 3.	0x00000064
0X0073	*BOARD_TLIM*	Board over temperature protection.	0x004B0046

MEMORY REGISTERS

ADDRESS	Name	Description	Reset
0X0080	*MEMORY_STAT*	Status of the on-board flash memory (memory state / usage information).	0x00000000
0X0081	*MEMORY_SIZE*	The size of memory available to write to.	0x00000000
0X0083-0X008A	*MEM_HLTH*	Memory health information registers.	0x00000000
0X0090	*MEM_WR_CTRL*	Memory Write interface control register.	0x00000000
0X0091	*MEM_WR_INDEX*	Start memory address of write operation (Index increments in 16KB clusters).	0x00000000
0X0092-0X0093	*MEM_WR_LEN*	Number of bytes (Lower 4 bytes) to write into memory.	0x00000000
0X0094	*MEM_WR_STAT*	Status of Write interface.	0x00000000
0X0095-0X0096	*MEM_WR_CNT*	Number of bytes written into memory (Lower 4 bytes).	0x00000000
0X0097	*MEM_WR_PKT_CNT*	Number of packets written into memory.	0x00000000
0X0098	*MEM_WR_ERR*	Error information field.	0x00000000
0X00A0	*MEM_RD_CTRL*	Memory Read interface control register.	0x00000000
0X00A1	*MEM_RD_INDEX*	Start memory address of read operation (Address increments in 16KB clusters).	0x00000000
0X00A2	*MEM_RD_LEN*	Number of packets to read from memory.	0x00000000
0X00A3	*MEM_RD_STAT*	Status of read interface.	0x00000000
0X00A4	MEM_RD_CNT	Packet count read from memory.	0x00000000
0X00A5	*MEM_RD_ERR*	Error information field.	0x00000000
0X00B0	*MEM_RT_CTRL*	Memory Retransmit interface control register.	0x00000000
0X00B1	*MEM_RT_QUEUE*	Retransmit queue register (Retransmit Index and Length).	0x00000000
0X00B2	*MEM_RT_STAT*	Status of retransmit interface.	0x00000000
0X00B4	*MEM_RT_CNT*	Number of retransmit packets read from memory in current/last active retransmit session.	0x00000000
0X00B5	*MEM_RT_ERR*	Error information field.	0x00000000

DATA ROUTING REGISTERS

ADDRESS	Name	Description	Reset
0X0110	*TF_CTRL*	Transfer frame control field.	0x00000002
0X0111	*TF_BUF_CNT*	Number of bytes in transfer frame buffer. Buffer holds incoming data until a frames length of bytes is available to create a data frame.	0x00000000
0X0112	*TF_CONF*	SCID, VCID, RT_VCID.	0x00002006
0X0113	*TF_DATA_CNT*	Transfer frame data count.	0x00000000
0X0114	*TF_OID_CNT*	Transfer frame idle count.	0x00000000
0X0115	*TF_STREAM_CNT*	Transfer frame stream count.	0x00000000
0X0120	*CROSSBAR*	Controls the flow/routing of the data in the transmitter	0x00000000
0X0121	*TP_GEN_CTRL*	Test pattern generator control.	0x00000000
0X0122	*TP_GEN_CNT*	Number of bytes generated by TP generator.	0x00000000
0X0123	*TP_GEN_RATE*	Data rate of generated test pattern data.	0x00000000
0X0124	*TP_CHK_CTRL*	Test pattern checker control.	0x00000000
0X0125	*TP_CHK_STAT*	Status of pattern checker.	0x00000000
0X0126	*TP_CHK_SYNC_C*NT	Number of bytes received while synced to a test pattern.	0x00000000
0X0127	*TP_CHK_UNSYNC_CNT*	Number of bytes received while not synced to a test pattern.	0x00000000
0X0128	*TP_CHK_RATE*	Data rate of received data.	0x00000000
0X0129	*TP_CHK_LST_CNT*	Number of bytes received at time of EOF signal reception.	0x00000000

PA STATUS AND CONTROL REGISTERS

ADDRESS	Name	Description	Reset
0X0190, 0X0191	*PA_STAT*	Flags showing overtemperature events, temperature, measured RF Power, measured current, and measured RF power quanta.	0x00000000
0X0192, 0X0193	*PA_CONF*	Target RF Power and PID mode control of the PA.	-
0X0197	*PA_LIM*	PA protection limits.	0x005A01C2
0X0198	*PA_TIMEOUT*	PA Timeout compare and current values.	0x04380000

9.2 REGISTER MAP

The subsections below provide a detailed description of all registers. Fields marked with a default value of ‘OS’ are configurable via an option sheet upon ordering.

All registers are accessable in all modes unless otherwise indicated.

9.2.1 GENERAL REGISTERS

9.2.1.1 ID

This register contains a fixed value identifying the VestaLink S25.

Register Address	0x0001
Default Value	0x53545832

Name	Bits	Access	Default	Description
ID	31-0	R	0x53545832	The fixed value is the ASCII representation of "STX2" in hexadecimal format.

9.2.1.2 SERIAL

This register contains the transmitter serial number.

Register Address	0x0002
Default Value	-

Name	Bits	Access	Default	Description
Serial	31-0	R	-	The serial number is unique for each VestaLink S25.

9.2.1.3 SW_VERSION, FW_VERSION, HW_VERSION

These registers contain the version numbers for the Software, Firmware and Hardware.

Register Address	0x0003, 0x0004, 0x0005
Default Value	-

Name	Bits	Access	Default	Description
Major [7:0]	7-0	R	-	Major version of the Software/Firmware/Hardware
Minor [7:0]	15-8	R	-	Minor version of the Software/Firmware/Hardware.
Patch [7:0]	23-16	R	-	Patch version of the Software/Firmware/Hardware.
Reserved	31-24	R	0	-

9.2.1.4 UPTIME

This register records the time that has elapsed since the transmitter has been powered on. This register can be read and checked against previous values to ensure the communication link to the radio is functional.

Register Address	0x0006
Default Value	0x00000000

Name	Bits	Access	Default	Description
Uptime	31-0	R	0	The time is captured in milliseconds.

9.2.1.5 SCRATCHPAD

This register can be written to verify that reading/writing is functional.

Register Address	0x0007
Default Value	0x00000000

Name	Bits	Access	Default	Description
Scratchpad	31-0	RW	0	The last written 32-bit message is stored to be read or overwritten.

9.2.2 COMMUNICATION REGISTERS

9.2.2.1 I2C_STAT

This register contains the status of I2C communication link. If any flags in this register are set, the I2C status flag in the BOARD_STAT register will also be set. This register is cleared on read.

Register Address	0x0010
Default Value	0x00000000

Name	Bits	Access	Default	Description
Invalid_Address	0	R	0	The user has attempted to write to an invalid address.
Out_Of_Range	1	R	0	The user has sent a read/write message of incorrect size.
Not_Writeable	2	R	0	The user has attempted to write a value to a read only register.
Internal_Err	3	R	0	An internal error has occurred.
Buffer_Overflow	4	R	0	Message queue buffer overflow occurred.
Reserved	31-5	R	0	-

9.2.2.2 I2C_CONF

This register contains the I2C address for the transmitter.

The unlock code must first be written to the Comm_Unlock register to be able to alter this register.

Register Address	0x0011
Default Value	-

Name	Bits	Access	Default	Description
Address [7:0]	7-0	RW	OS	The address for the I2C communication link. The address will be reset to default when the transmitter is power cycled.
Reserved	31-8	R	0	-

9.2.2.3 CAN_STAT

This register contains the status of the CAN communication links. If any flags in this register are set, the relavant CAN status flag in the BOARD_STAT register will also be set. This register is cleared on read.

Register Address	0x0012, 0x0016
Default Value	0x00000000

Name	Bits	Access	Default	Description
Invalid_Address	0	R	0	The user has attempted to write to an invalid address.
Out_Of_Range	1	R	0	The user has sent a read/write message of incorrect size.
Not_Writeable	2	R	0	The user has attempted to write a value to a read only register.
Internal_Err	3	R	0	An internal error has occurred.
Buffer_Overflow	4	R	0	Message queue buffer overflow occurred.
Reserved	31-5	R	0	-

9.2.2.4 CAN_CONF

These registers contain the CAN addresses and communication protocols for the transmitter.

The unlock code must first be written to the Comm_Unlock register to be able to alter this register.

Different protocols are not currently supported on CAN0 and CAN1, respectively (both must be configured to the same protocol and is enforced as such).

Register Address	0x0013, 0x0017
Default Value	-

Name	Bits	Access	Default	Description
Address [15:0]	15-0	RW	OS	The address for the CAN communication link. The address will be reset to default when the transmitter is power cycled.
Protocol [3:0]	19-16	RW	OS	The protocol used for the CAN communication link. **CSPv2 & CAN-TS protocols are only supported from software v2.1.0 onward.**
Reserved	31-20	R	0	-

Table 9-2: Protocol Options

Protocol	Value
Cubecom Protocol	0x00
Cubesat Space Protocol (CSP) V2	0x01
CAN-TS	0x02

9.2.2.5 CAN_CSP_CONF

This CSP configuration for the transmitter**.**

The unlock code must first be written to the Comm_Unlock register to be able to alter this register.

These registers are only applicable when the communication protocol has been set to CSP.

Register Address	0x0326, 0x0327
Default Value	-

Name	Bits	Access	Default	Description
Mask [3:0]	3-0	RW	OS	Network mask to use for CSP packets.
IsDefault	4	RW	OS	Whether or not the CAN interface is reachable by default.
Reserved	31-5	R	0	-

9.2.2.6 CANTS_CONF

This register CAN-TS configuration for the transmitter.

The unlock code must first be written to the Comm_Unlock register to be able to alter this register.

This register is only applicable when the communicaiton protocol has been set to CAN-TS.

Register Address	0x0328
Default Value	-

Name	Bits	Access	Default	Description
Keep-alive interval [15:0]	15-0	RW	OS	Interval in milliseconds at which CAN-TS keepalive messages are transmitted on the primary CAN interface.
Primary interface [1:0]	17-16	RW	OS	Primary CAN interface to use for CAN-TS communicaiton. This will determine the interface on which keepalive messages are broadcasted. When valid CAN-TS packets are received and are targetted at the transmitter, it will automatically update the primary interface to that interface. **This is only applicable to VestaLink S25 hardware revisions C0 and onward. Prior versions only had a single CAN interface available.**
Reserved	31-18	R	0	-

Table 9-3: Interface Options

Interface	Value
CAN0	0x00
CAN1	0x01

9.2.2.7 UART_STAT

This register contains the status of the UART communication link. If any flags in this register are set, the UART status flag in the BOARD_STAT register will also be set. This register is cleared on read.

Register Address	0x0014
Default Value	0x00000000

Name	Bits	Access	Default	Description
Invalid_Address	0	R	0	The user has attempted to write to an invalid address.
Out_Of_Range	1	R	0	The user has sent a read/write message of incorrect size.
Not_Writeable	2	R	0	The user has attempted to write a value to a read only register.
Internal_Err	3	R	0	An internal error has occurred.
Buffer_Overflow	4	R	0	Message queue buffer overflow occurred.
Reserved	31-5	R	0	-

9.2.2.8 UART_CONF

This register contains the UART address for the transmitter. The UART interface is configured to a baud rate of 115200.

The unlock code must first be written to the Comm_Unlock register to be able to alter this register.

Register Address	0x0015
Default Value	-

Name	Bits	Access	Default	Description
Address [7:0]	7-0	RW	OS	The address for the UART communication link. The address will be reset to default when the transmitter is power cycled.
Reserved	31-8	R	0	-

9.2.2.9 COMM_UNLOCK

This register is used to unlock the transmitter communication configuration registers. The unlock code is **0x15D3**.

Register Address	0x001F
Default Value	0x00000000

Name	Bits	Access	Default	Description
Unlock_Code [15:0]	15-0	RW	0x0000	Writing the unlock code to the register will unlock the communication config registers. Writing anything else to it will lock the communication registers.
Status	16	R	0	Indicates if in locked (‘0’) or unlocked (‘1’) state.
Reserved	31-17	R	0	-

9.2.3 DATA SOURCE REGISTERS

9.2.3.1 HS_NODE_INFO

These registers contain fixed values to determine the type of data interface that the node has been configured as. The data interface referred to as “Node 0” (Interface 0 on J2) is represented by the registers ranged 0x0020 to 0x0025 while the data interface referred to as “Node 1” (Interface 1 on J3) is represented by the registers ranged 0x0026 to 0x002B.

Register Address	0x0020/0x0026
Default Value	-

Name	Bits	Access	Default	Description
Node_Info	31-0	R	-	The fixed value is the ASCII representation of the type of data link in hexadecimal format.

Table 9-4: High Speed Interface Data Link Types.

Value	ASCII Representation	Description
0x00000000		Pre Flash Access/Config enter value
0x53505730	SPW0	SpaceWire Node 0
0x53505731	SPW1	SpaceWire Node 1
0x4C525830	LRX0	CUBECOM LVDS Receiver Node 0
0x4C525831	LRX1	CUBECOM LVDS Receiver Node 1
0x53504930	SPI0	SPI Node 0
0x53504931	SPI1	SPI Node 1

9.2.3.2 HS_CTRL

This register contains the control bits used to operate the configured data link.

These registers can only be written in Flash-Access, Config, and Transmit modes.

Register Address	0x0021/0x0027
Default Value	0x00000000

SpaceWire				
Name	**Bits**	**Access**	**Default**	**Description**
Auto_Start	0	RW	0	The Auto_Start bit puts the SpaceWire interface in a passive connection mode where the connection is initiated from the source device.
Link_Start	1	RW	0	The Link_Start bit configures the SpaceWire interface to actively initiate a connection with the source.
Link_Disable	2	RW	0	The Link_Disable bit is used to disable the SpaceWire core to save power.
Ignore_EOP	3	RW	0	By default, SpaceWire EOP control characters are passed through the data chain as an EOF characters (this will cuase memory write operations to be completed). Setting this flag will disable this behaviour.
Clr_Rx_Buf	4	RW	0	Clear receive buffer (64 bytes).
Clr_Tx_Buf	5	RW	0	Clear transmit buffer (64 bytes).
Reserved	7-6	R	0	-
Tx_Clk_Div [7:0]	15-8	RW	0	This register sets the SpaceWire transmit bit rate to the source. The transmit bit rate becomes active once a link has been established. The bit rate is calculated as follows: TxBitRate = Tx_Clk / (TX_DIV+1), where TxBitRate is the transmit bit rate in Mbps. (Tx_Clk = 200)
Reserved	31-16	R	0	-

LVDS Receiver				
Name	**Bits**	**Access**	**Default**	**Description**
Enable	0	RW	0	Enables the LVDS interface. The status registers and counts are reset when the link is disabled.
Channel [1:0]	2-1	RW	0	The channel configuration determines the data lines used. Configuration shown in Table 9-5.
Mode [1:0]	4-3	RW	0	The mode configuration determines the data latching relative to the LVDS clock. Configuration shown in Table 9-6.
Ignore_Parity	5	RW	0	Ignores the parity check when set to ‘1’.
Reserved	31-6	R	0	-

Table 9-5: LVDS channel configuration.

Value	Channel Config
0	Only D0 used
1	Only D1 used
2	Both D0 and D1 are used
3	Reserved

Table 9-6: LVDS mode configuration.

Value	Mode Config
0	Rising-edge latching (SDR)
1	Falling-edge latching (SDR)
2	Rising and falling-edge latching (DDR)
3	Reserved

SPI Receiver				
Name	**Bits**	**Access**	**Default**	**Description**
Enable	0	RW	0	Enables the SPI data link. The counters and status flags are reset when the link is disabled.
Mode [1:0]	2-1	RW	0	Indicates to the SPI link which SPI mode the system must be configured to. Mode change is only accepted when (re)enabling the SPI interface.
Ready_Polarity	3	RW	0	Changes if a '1' or '0' indicates if transmitter is ready to receive SPI data. If Ready_Polarity is '0' a '1' on the ready line indicates the system is ready for data and vice versa.
Reserved	31-4	R	0	-

9.2.3.3 HS_STAT_0

These registers contain the status of the configured data interfaces. Status bits are cleared on a register read (Unless otherwise stated).

Register Address	0x0022/0x0028
Default Value	0x00000000

SpaceWire				
Name	**Bits**	**Access**	**Default**	**Description**
Conn_State [1:0]	1-0	R	0b00	These two bits indicate the connection state of the SpaceWire link. This is the link's current state. Connections states listed in Table 9-7.
Rx_Recved	2	R	0	A byte or more has been received on the SpaceWire link.
Tx_Sent	3	R	0	A byte or more has been transmitted on the SpaceWire link.
EEP_Detect	4	R	0	An Error End of Packet (EEP) control character was received.
EOP_Detect	5	R	0	An End of Packet (EOP) control character was received.
Err_Detect	6	R	0	An error was detected in the SpaceWire link from the last time the register was read.
Cred_Err	7	R	0	Credit error occurred.
Esc_Err	8	R	0	Escape error occurred.
Par_Err	9	R	0	Parity error occurred.
Disc_Err	10	R	0	Disconnect error occurred.
Reserved	15-11	R	0	-
Rx_Buf_Cnt [7:0]	23-16	R	0	Receive buffer byte count (Buffer count 64 bytes).
Tx_Buf_Cnt [7:0]	31-24	R	0	Transmit buffer byte count (Buffer count 64 bytes).

Table 9-7: SpW connection status values.

Value	Description
0b00	Link off
0b01	Ready for connection
0b10	Link connecting
0b11	Link established

LVDS Receiver				
Name	**Bits**	**Access**	**Default**	**Description**
Active_Clk	0	R	0	Indicates that an active clock is currently running on the LVDS_CLK line.
Synced	1	R	0	When several consecutive sync words (0x03EA) have been successfully received the link will move to a synced state. Data can be sent in the synced state. The flow control line LVDS_FC will also indicate the system is synced and ready to receive data.
Sync_Recved	2	R	0	Indicates that a sync word has been received.
Byte_Recved	3	R	0	Indicates that a byte of valid data has been received.
Cmd_Recved	4	R	0	Indicates that a valid command has been received.
Unk_Recved	5	R	0	Indicates that an unknown word has been received.
Byte_Ignored	6	R	0	If the flow control line is low while the system is still synced (Indicating that the LVDS buffer is full) any word sent across will be ignored.
Parity_Error	7	R	0	Indicates a parity error has occurred resulting in the link to move to an unsynchronized state.
Link_Oversat	8	R	0	The LVDS link is being run faster than the receive node can receive and decode the data.
Reserved	12-9	R	0	-
Buffer_Cnt [9:0]	23-13	R	0x00	The Byte count of the received data bytes (Buffer size is 1024 bytes).
Sync_Cnt [7:0]	31-24	R	0x00	Running 8-bit counter of received sync words after a link has been established.

SPI Receiver				
Name	**Bits**	**Access**	**Default**	**Description**
Buffer_Full	0	R	0	Indicates the buffer is full. Overflow will occur if more data is sent over the SPI interface.
Buffer_Overflow	1	R	0	Indicates data was received while the buffer is full. The sent/received data was not captured. Data has been lost.
Buffer_Ready	2	R	0	Indicates the level of the flowcontrol of the interface. If the flag is ‘1’ the receive buffer level is below the halfway point (less than 512 bytes) and the flowcontrol signal indicates the interface is ready for more data. If the flag is ‘0’ the receive buffer level is above the halfway point (more than 512 bytes) and the flowcontrol signal indicates the interface is not ready for more data.
Rx_Recved	3	R	0	Data was received on the SPI interface.
Reserved	15-4	R	0	-
Buffer_Count [15:0]	31-16	R	0x0000	The buffer count indicates the number of 32-bit words that have been captured in the 1024-word (4096 bytes) buffer.

9.2.3.4 HS_STAT_1

These registers contain the statuses of the configured data interfaces.

Register Address	0x0023/0x0029
Default Value	0x00000000

SpaceWire				
Name	**Bits**	**Access**	**Default**	**Description**
Cred_Err_Cnt [7:0]	7-0	R	0x00	This is an 8-bit register that counts credit errors in the SpaceWire interface.
Esc_Err_Cnt [7:0]	15-8	R	0x00	This is an 8-bit register that counts escape errors in the SpaceWire interface
Par_Err_Cnt [7:0]	23-16	R	0x00	This is an 8-bit register that counts parity errors in the SpaceWire interface.
Disc_Err_Cnt [7:0]	31-24	R	0x00	This is an 8-bit register that counts disconnect errors in the SpaceWire interface.

LVDS Receiver				
Name	**Bits**	**Access**	**Default**	**Description**
Cmd_Cnt [7:0]	7-0	R	0x00	Running 8-bit counter of received commands after a link has been established.
Unk_Cnt [7:0]	15-8	R	0x00	Running 8-bit counter of received unknown words after a link has been established.
Ignore_Cnt [7:0]	23-16	R	0x00	The number of data packets ignored. Data is ignored when the system is synchronized and the buffer is full. The flow control line indicates when the system is ready to receive data and will indicate the system is not ready when the buffer contains more than 100 bytes. The buffer has some room to accommodate data bytes sent after the flow control line goes low.
Parity_Cnt [7:0]	31-24	R	0x00	The number of times a parity error was received while the link has been enabled.

SPI Receiver				
Name	**Bits**	**Access**	**Default**	**Description**
Reserved	31-0	R	0	-

9.2.3.5 HS_RX_CNT

These registers contain byte counts of data received through the configured data interfaces.

Register Address	0x0024/0x002A
Default Value	0x00000000

Name	Bits	Access	Default	Description
Rx_Count	31-0	R	0	This is the 32-bit register that holds the data count of the received data bytes.

9.2.3.6 HS_TX_CNT

These registers contain byte counts of data transmitted through the configured data interfaces.

Register Address	0x0025/0x002B
Default Value	0x00000000

Name	Bits	Access	Default	Description
Tx_Count	31-0	R	0	This is the 32-bit register that holds the data count of the transmitted data bytes. Not applicable to LVDS and SPI interfaces since they are implemented as RX only.

9.2.3.7 SPI_CTRL

This register is used to configure and control the SPI data interface (PC104 connector).

This register can only be written in Flash-Access, Config, and Transmit modes.

Register Address	0x002C
Default Value	0x00000000

Name	Bits	Access	Default	Description
Enable	0	RW	0	Enables the SPI data link. The counters and status flags are reset when the link is disabled.
Mode [1:0]	2-1	RW	0	Indicates to the SPI link which SPI mode the system must be configured to. Mode change is only accepted when (re)enabling the SPI interface.
Ready_Polarity	3	RW	0	Changes if a '1' or '0' indicates if transmitter is ready to receive SPI data. If Ready_Polarity is '0' a '1' on the ready line indicates the system is ready for data and vice versa.
Reserved	31-4	R	0	-

9.2.3.8 SPI_STAT

This register is used to monitor the SPI data link status (PC104 connector). The counters and status flags are reset when the link is disabled.

Register Address	0x002D
Default Value	0x00000000

Name	Bits	Access	Default	Description
Buffer_Full	0	R	0	Indicates the buffer is full. Overflow will occur if more data is sent over the SPI interface.
Buffer_Overflow	1	R	0	Indicates data was received while the buffer is full. The sent/received data was not captured. Data has been lost.
Buffer_Ready	2	R	0	Indicates the level of the flowcontrol of the interface. If the flag is '1' the receive buffer level is below the halfway point (less than 512 bytes) and the flowcontrol signal indicates the interface is ready for more data. If the flag is '0' the receive buffer level is above the halfway point (more than 512 bytes) and the flowcontrol signal indicates the interface is not ready for more data.
Rx_Recved	3	R	0	Data received on the SPI interface.
Reserved	15-4	R	0	-
Buffer_Count [15:0]	31-16	R	0	The buffer count indicates the number of 32-bit words that have been captured in the 1024-word (4096 bytes) buffer.

9.2.3.9 SPI_RX_CNT

This register can be used to determine the number of data that is sent through the SPI interface.

Register Address	0x002E
Default Value	0x00000000

Name	Bits	Access	Default	Description
RX_Count	31-0	R	0	This is the 32-bit register that holds the data count through the SPI interface.

9.2.4 TELEMETRY REGISTERS

9.2.4.1 BOARD_STAT

This register contains a collection of flags indicating any issue or status in one of the other status registers.

Register Address	0x0030
Default Value	0x00000000

Name	Bits	Access	Default	Description
I2C_Status	0	R	0	This flag is set if any of the status flags are set in the *I2C_STAT* register. This flag is cleared when the I2C status register is read.
CAN0_Status	1	R	0	This flag is set if any of the status flags are set in the *CAN*_STAT register. This flag is cleared when the CAN0 status register is read.
UART_Status	2	R	0	This flag is set if any of the status flags are set in the *UART_STAT* register. This flag is cleared when the UART status register is read.
MEM_Status	3	R	0	This flag is set if the memory fails to initialise. Memory status can be found in further detail in Section 9.2.7. This flag is cleared when memory successfully initialises
RF_Status	4	R	0	This flag is set if the RF subsystem loses lock (*RF_STAT* register). Flag is cleared when re-entering transmit mode.
PA_Status	5	R	0	This flag is set if any of the status flags are set in the *PA_STA* register. Flag is cleared when re-entering transmit mode.
PA_Warn_Status	6	R	0	This flag is set when the PA temp exceeds the warn temp limit set in *PA_LIM*.
Timeout_Status	7	R	0	This flag is set when the transmitter has been transmitting for more than the set time limit. See Section 9.2.9.4 for more detail on this configuration. Flag is cleared when re-entering transmit mode.
Brd_Warn_Status	8	R	0	This flag is set when the board temperature exceeds the warn temperature limit set in *BOARD_TLIM*.
Brd_Trip_Status	9	R	0	This flag is set when the board temperature exceeds the trip temperature limit set in *BOARD_TLIM*. Flag is cleared when re-entering transmit mode.
Brd_Curr_Status	10	R	0	This flag is set when a current sensor is reporting an anomalous reading. See Section 9.2.4.4. This flag is cleared when the *CURRENT_ERR* register is read.
Brd_Volt_Status	11	R	0	This flag is set when a voltage sensor is reporting an anomalous reading. See Section 9.2.4.5. This flag is cleared when the *VOLTAGE_ERR* register is read.
CAN1_Status	12	R	0	This flag is set if any of the status flags are set in the *CAN*_STAT register. This flag is cleared when the CAN1 status register is read.
Reserved	31-13	R	0	-

9.2.4.2 RF_STAT

This register contains the RF status flags. The RF section is powered up when the transmitter enters config mode. When in config mode and the RF system has successfully been initialized then the RF components will indicate if they are locked. While in transmit mode if lock is lost the transmitter will jump back into config mode.

Register Address	0x0031
Default Value	0x00000000

Name	Bits	Access	Default	Description
RF_Synth	0	R	0	This bit indicates if the RF synthesizer is locked (1 indicates locked).
IF_Synth	1	R	0	This bit indicates if the IF synthesizer is locked (1 indicates locked).
Reserved	31-2	R	0	-

9.2.4.3 TEMP

These registers contain temperature measurements for the VestaLink S25. The values are 16-bit signed values, in 0.25°C increments. A value of 100 implies a temperature of 25°C.

Register Address	0x0032 – 0x0035
Default Value	0x00000000

Name	Bits	Access	Default	Description
				TEMP_0 (0x0032)
PA_Temp [15:0]	15-0	R	0	The temperature of the PA is determined by measuring the top of the case temperature. This is closely related to the junction temperature of the PA. This is a 16-bit signed value, in 0.25°C increments. A value of 100 implies a temperature of 25°C.
PA_SMPS_Temp [15:0]	31-16	R	0	The temperature of the PA DC-DC converter is measured just adjacent to the switcher. This is a 16-bit signed value, in 0.25°C increments. A value of 100 implies a temperature of 25°C.
				TEMP_1 (0x0033)
Board_Temp_0 [15:0]	15-0	R	0	The board temperature is measured by the PA controller which has an internal temperature sensor. This gives a fairly accurate representation of the PCB temperature.
FPGA_Int_Temp [15:0]	31-16	R	0	The temperature of the FPGA, as reported by its internal temperature sensor.
				TEMP_2 (0x0034)
Board_Temp_1 [15:0]	15-0	R	0	This temperature measurement is made by a voltage and current monitoring sensor located on the board.
Board_Temp_2 [15:0]	31-16	R	0	This temperature measurement is made by a voltage and current monitoring sensor located on the board.
				TEMP_3 (0x0035)
Board_Temp_3 [15:0]	15-0	R	0	This temperature measurement is made by a voltage and current monitoring sensor located close to the PA SMPS.
Reserved	31-16	R	0	-

9.2.4.4 CURRENT_ERR

This register is the aggregate of all the current sensor's warning flags in one register. The register is reset when read.

Register Address	0x003E
Default Value	0x00000000

Name	Bits	Access	Default	Description
PA_VBAT	0	R	0	A value of '1' for any bit implies that an anomalous current level was detected. See Section 8.6 for more detail on the power network.
PA_VCC	1	R	0	
Board_VBAT	2	R	0	
FPGA_1V0	3	R	0	
FPGA_1V8	4	R	0	
FPGA_2V5	5	R	0	
FPGA_3V3	6	R	0	
Board_3V6	7	R	0	
Board_5V3	8	R	0	
-	9	R	0	
-	10	R	0	
-	11	R	0	
S_Drv_Amp_5V	12	R	0	
IF_Synth_3V3	13	R	0	
-	14	R	0	
-	15	R	0	
IF_VCP_5V	16	R	0	
Power_Ctrl_5V	17	R	0	
Mod_3V3	18	R	0	
Var_Filt_5V	19	R	0	
Mem_1V8	20	R	0	
Mem_3V3	21	R	0	
C_Dist_3V3	22	R	0	
C_Ref_3V3	23	R	0	
DAC_3V3	24	R	0	
Reserved	31-25	R	0	-

9.2.4.5 VOLTAGE_ERR

This register is the aggregate of all the voltage sensor's warning flags in one register. The register is reset to when read.

Register Address	0x003F
Default Value	0x00000000

Name	Bits	Access	Default	Description
PA_VBAT	0	R	0	A value of '1' for any bit implies that an anomalous voltage level was detected. See Section 8.6 for more detail on the power network.
PA_VCC	1	R	0	
Board_VBAT	2	R	0	
FPGA_1V0	3	R	0	
FPGA_1V8	4	R	0	
FPGA_2V5	5	R	0	
FPGA_3V3	6	R	0	
Board_3V6	7	R	0	
Board_5V3	8	R	0	
-	9	R	0	
-	10	R	0	
-	11	R	0	
S_Drv_Amp_5V	12	R	0	
IF_Synth_3V3	13	R	0	
-	14	R	0	
-	15	R	0	
IF_VCP_5V	16	R	0	
Power_Ctrl_5V	17	R	0	
Mod_3V3	18	R	0	
Var_Filt_5V	19	R	0	
Mem_1V8	20	R	0	
Mem_3V3	21	R	0	
C_Dist_3V3	22	R	0	
C_Ref_3V3	23	R	0	
DAC_3V3	24	R	0	
Reserved	31-25	R	0	-

9.2.4.6 CURRENTS

These registers contain current measurements for the VestaLink S25. The sensor numbers align with their bit positions in the current error register.

Register Address	0x0040-0x004F
Default Value	0x00000000

Name	Bits	Access	Default	Description
Sensor N [15:0]	15-0	R	0	This is a 16-bit number measured in 1 mA increments. The current sensor registers can be found in Table 9-8.
Sensor N+1 [15:0]	31-16	R	0	

Table 9-8: Current Sensor Registers

Register Address	Sensor N		Sensor N+1	
	Sense No.	Name	Sense No.	Name
0x0040	0	PA_VBAT	1	PA_VCC
0x0041	2	Board_VBAT	3	FPGA_1V0
0x0042	4	FPGA_1V8	5	FPGA_2V5
0x0043	6	FPGA_3V3	7	Board_3V6
0x0044	8	Board_5V3	9	-
0x0045	10	-	11	-
0x0046	12	S_Drv_Amp_5V	13	IF_Synth_3V3
0x0047	14	-	15	-
0x0048	16	IF_VCP_5V	17	Power_Ctrl_5V
0x0049	18	Mod_3V3	19	Var_Filt_5V
0x004A	20	Mem_1V8	21	Mem_3V3
0x004B	22	C_Dist_3V3	23	C_Ref_3V3
0x004C	24	DAC_3V3	25	-
0x004D	*26	PA_0	27	-
0x004E	28	-	29	-
0x004F	30	-	31	-

Note:

*26 - Current measurement of PA_0 measured using PA power controller more details in Section 9.2.9.

9.2.4.7 VOLTAGES

This register contains voltage measurements for the VestaLink S25. The sensor numbers align with their bit positions in the voltage error register.

Register Address	0x0050-0x005F
Default Value	0x00000000

Name	Bits	Access	Default	Description
Sensor N [15:0]	15-0	R	0	This is a 16-bit number measured in mV. The voltage sensor registers can be found in Table 9-8.
Sensor N+1 [15:0]	31-16	R	0	

Table 9-9: Voltage Sensor Registers

Register Address	Sensor N		Sensor N+1	
	Sense No.	Name	Sense No.	Name
0x0050	0	PA_VBAT	1	PA_VCC
0x0051	2	Board_VBAT	3	FPGA_1V0
0x0052	4	FPGA_1V8	5	FPGA_2V5
0x0053	6	FPGA_3V3	7	Board_3V6
0x0054	8	Board_5V3	9	-
0x0055	10	-	11	-
0x0056	12	S_Drv_Amp_5V	13	IF_Synth_3V3
0x0057	14	-	15	-
0x0058	16	IF_VCP_5V	17	Power_Ctrl_5V
0x0059	18	Mod_3V3	19	Var_Filt_5V
0x005A	20	Mem_1V8	21	Mem_3V3
0x005B	22	C_Dist_3V3	23	C_Ref_3V3
0x005C	24	DAC_3V3	25	-
0x005D	*26	FPGA_Int_1V	*27	FPGA_Int_1V8
0x005E	*28	FPGA_Int_2V5	29	-
0x005F	30	-	31	-

Note: *26-28 - Voltage measurements are from the FPGA internal sensor.

9.2.5 CONTROL REGISTERS

9.2.5.1 MODE

This register sets the operating mode of the VestaLink S25.

Register Address	0x0060
Default Value	0x00000000

Name	Bits	Access	Default	Description
Mode [7:0]	7-0	RW	0	Set the current (desired) mode of the VestaLink S25. When this register is written to the transmitter will start going to the desired mode. When it reaches that mode, this register will reflect it. Valid settings are given in Table 9-10. Writes to this field is not processed when a mode change is alread in progress ("Busy" flag below is set).
Busy	8	R	0	This bit indicates the VestaLink S25 is busy changing mode.
Error	9	R	0	This bit indicates that there was an error during the mode change.
Reserved	31-10	R	0	-

Table 9-10: Transmitter Operating States

Name	Value	Description
Standby	0x00	Lowest power mode. Board telemetry is active.
Flash Access	0x01	Only memory enabled. Access mode if the payload data is to be stored to the on-board flash memory.
Config	0x02	All the peripherals and RF subsystems, except for the PA, are enabled.
Transmit	0x03	The PA is turned on and the RF signal is transmitted.

9.2.5.2 SYMBOL_RATE

This register sets the symbol rate of the transmitter.

Register Address	0x0061
Default Value	-

Name	Bits	Access	Default	Description
Symbol_Rate	31-0	RW	OS	The unit for the symbol rate is in ksps. Set the register to 5000 ksps for a symbol rate of 5 Msps.

9.2.5.3 FREQUENCY

This register sets the frequency of the transmitter.

Register Address	0x0062
Default Value	-

Name	Bits	Access	Default	Description
Frequency	31-0	RW	OS	The frequency is in kHz. The frequency steps in 1 MHz steps. Default of 0x00224188 sets frequency to 2245 MHz. Step changes lower than 1 MHz are ignored.

9.2.5.4 ENCODING_CTRL

This register sets the universal encoding parameters.

Register Address	0x0065
Default Value	-

Name	Bits	Access	Default	Description
Encoding	0	RW	OS	Sets which encoding method is used. **This field cannot be modified when in Transmit mode.**
RRC [1:0]	2-1	RW	0x02	Sets the Rolloff for the encoding. See Table 9-12 for RRC options.
Insert_ASM	3	RW	1	Enables or disables ASM insertion. DVB-S2 = ASM inserted in front of transfer frame. CCSDS = ASM used inside encoding core to indicate start of encoding frame.
Invert_I	4	RW	0	Inverts the sign of the I samples. ('1' = enabled)
Invert_Q	5	RW	0	Inverts the sign of the Q samples. ('1' = enabled)
Swap_IQ	6	RW	0	Swaps the I and Q samples. ('1' = enabled)
Reserved	15-7	R	0	-
Frame_Length [15:0]	31-16	RW	-	Frame length is the data length of the user data inside the transfer frames (Dependent on active encoding and encoding settings shown in Table 9-13).

Table 9-11: Encoding options

Encoding	Value
DVB-S2	0x00
CCSDS	0x01

Table 9-12: RRC Options

Rolloff (RRC)	Value
0.35	0x00
0.25	0x01
0.2	0x02

Table 9-13: Frame Lengths

Setting	Frame Length Constraint
DVB-S2	200 <= Len <= 16380 (Recommended length is 16380 to align memory packets)
CCSDS with RS_239/255 selected	Constant 233 (239 – 6 (header bytes) = 233)
CCSDS with RS_223/255 selected	Constant 217 (223 – 6 (header bytes) = 217)
CCSDS without Reed-Solomon encoding	200 <= Len <=2042

9.2.5.5 DVBS2_CONF

This register sets the DVB-S2 encoding. The MODCODs are typically changed during data transmission, while the rest of the fields are kept constant for a mission.

Register Address	0x0066
Default Value	0x00000014

Name	Bits	Access	Default	Description
MODCOD [7:0]	7-0	RW	4	Sets the modulation and coding used for the DVB-S2 standard. See Table 9-14 for possible MODCOD values. This table has been re-created from the DVB-S2 standards document (ETSI EN 302 307). Only bits 4-0 are used.
Pilot_En	8	RW	1	Enables or disables pilot insertion for DVB-S2. Set to '1' to enable, or '0' to disable.
FEC_Frame	9	RW	0	Set the DVB-S2 LDPC encoding frame size to normal (0) or short (1) frames. Normal frames have a length of 64800 bits and 16200 bits for short frames. *Note: If short frames are selected, 9/10 code rates are not supported for short frames per the DVB-S2 standard.
Reserved	31-10	R	0	-

Table 9-14: MODCOD Options

Coding	Modulation			
	QPSK	8PSK	16APSK	32APSK
1/4	1	-	-	-
1/3	2	-	-	-
2/5	3	-	-	-
1/2	4	-	-	-
3/5	5	12	-	-
2/3	6	13	18	-
3/4	7	14	19	24
4/5	8	-	20	25
5/6	9	15	21	26
8/9	10	16	22	27
9/10	11	17	23	28

9.2.5.6 CCSDS_CONF

This register sets the CCSDS encoding. Changes to these parameters while data is transmitted will result in data loss.

Register Address	0x0067
Default Value	0x0001002A

Name	Bits	Access	Default	Description
Modulation	1-0	RW	2	Sets the modulation used for the CCSDS standard. (Available modulation schemes shown in Table 9-15)
Randomizer	2	RW	0	Enables the randomizer
RS_Enable	3	RW	1	Enables the Reed-Solomon encoder in the path
RS_Code	4	RW	0	'0' = RS 239/255 '1' = RS 223/255
RS_Dual2Conv	5	RW	1	Enables the dual basis to conventional converter. ('1' enables the converter, '0' bypasses it)
RS_Interleave	8-6	RW	0	Sets the amount of interleaving in the RS encoding.(Interleaver options shown in Table 9-16)
Reserved	15-9	R	0	-
Conv_Enable	16	RW	1	Enables the convolutional encoder in the path.
Conv_KR	19-17	RW	0	Sets the coding rate of the convolutional encoder. (Code rate shown in Table 9-17)
Conv_Diff	20	RW	0	Enables the differential encoding of the convolutional encoder. (Converts from NRZ-L to NRZ-M)
Reserved	31-21	R	0	-

Table 9-15: Modulation Options

Modulation	Value
BPSK	0x00
QPSK	0x01
OQPSK	0x02

Table 9-16: Interleaving

Interleaving	Value
1 (No Interleaving)	0x00
2	0x01
3	0x02
4	0x03
5	0x04
8	0x05

Table 9-17: Convolutional Code Rate

Coding Rate	Value
1/2	0x00
2/3	0x01
3/4	0x02
5/6	0x03
7/8	0x04

9.2.5.7 ENCODING_STAT

This register contains the number of dummy frames generated by the DVB-S2 core.

Register Address	0x0068
Default Value	0x00000000

Name	Bits	Access	Default	Description
Dummy_Count	31-0	R	0	When no data is available into the core the core will generate Dummy Frames to keep the line active. The dummy count register keeps count of the generated dummy frames. The count will wrap back to 0 if incrementing past 0xFFFFFFFF.

9.2.6 THERMAL PROTECTION REGISTERS

9.2.6.1 PROTECTION_LEVEL_1

The first level of PA over temperature protection. Monitored temperature is PA_Temp discussed in Section 9.2.9.1.

Register Address	0x0070
Default Value	0x001E005F

Name	Bits	Access	Default	Description
Temperature [15:0]	15-0	RW	95	The temperature limit that will trigger the protection procedure when the PA temperature exceeds it. Temperature in (°C). Set temperature 255 to ignore this level.
Power [15:0]	31-16	RW	30	RF Power to throttle the transmitter to if the limit is triggered. Power is in dBm. Setting power to 0 will set the board back to Standby mode if the limit is triggered.

9.2.6.2 PROTECTION_LEVEL_2

The second level of PA over temperature protection. Monitored temperature is PA_Temp discussed in Section 9.2.9.1.

Register Address	0x0071
Default Value	0x001B0061

Name	Bits	Access	Default	Description
Temperature [15:0]	15-0	RW	97	The temperature limit that will trigger the protection procedure when the PA temperature exceeds it. Temperature in (°C). Set temperature 255 to ignore this level.
Power [15:0]	31-16	RW	27	RF Power to throttle the transmitter to if the limit is triggered. Power is in dBm. Setting power to 0 will set the board back to Standby mode if the limit is triggered.

9.2.6.3 PROTECTION_LEVEL_3

The third level of PA overtemperature protection. Monitored temperature is PA_Temp discussed in Section 9.2.9.1.

Register Address	0x0072
Default Value	0x00000064

Name	Bits	Access	Default	Description
Temperature [15:0]	15-0	RW	100	The temperature limit that will trigger the protection procedure when the PA temperature exceeds it. Temperature in (°C). Set temperature 255 to ignore this level.
Power [15:0]	31-16	RW	0	RF Power to throttle the transmitter to if the limit is triggered. Power in dBm. Setting power to 0 will set the board back to Standby mode if the limit is triggered.

9.2.6.4 BOARD_TLIM

The temperature limit of the board. Monitored temperature is Board_Temp_0 discussed in Section 9.2.4.3

Register Address	0x0073
Default Value	0x004B0046

Name	Bits	Access	Default	Description
Temp_Warn [15:0]	15-0	RW	70	The temperature limit that will trigger the temperature warning if the temperature exceeds this limit.
Temp_Lim [15:0]	31-16	RW	75	The temperature limit that will trigger the board to go back to Standby mode if the temperature exceeds this limit.

9.2.7 MEMORY REGISTERS

None of these registers can be modified in Idle mode, since the memory is not enabled in this mode.

9.2.7.1 MEMORY_STAT

This register contains the status information of the on-board flash memory (The status information is captured on each memory initialization step).

*Note: Writing any value to this register updates the values in this register and the other memory health registers.

Register Address	0x0080
Default Value	0x00000000

Name	Bits	Access	Default	Description
Memory_State [7:0]	7-0	R	0	The state of the flash memory is described in Table 9-18.
EOL_Info [7:0]	15-8	R	0	Indicates memory end of life information reflected by average reserved blocks consumed. Further details in Table 9-19.
Usage_Est [7:0]	23-16	R	0	Indicates the estimated memory lifetime usage determined by wear estimates (See Table 9-20).
Reserved	31-24	R	0	-

Table 9-18: Flash Memory state

Memory State Value	Memory State	Description
0	Memory Off	The memory is off (Currently in Standby Mode).
1	Memory Idle	The memory has been powered and correctly initialised. The memory is in a state where it can be interacted with.
2	Write Busy	A memory write process is currently ongoing and must be stopped before another process can be started.
3	Read Busy	A memory read process is currently ongoing and must be stopped before another process can be started (Except for a retransmit process).
4	Retransmit Busy	A memory retransmit process is currently ongoing and must be stopped before another process can be started (Except for a read process).
5	Health Update Busy	The memory status is currently being read from the memory. No other process can be started.
6	Init Failed	The memory failed to initialise on power up. Switch back to Standby and then Flash Access/Config mode to try reinitialising the memory.
7	Interface Error	The physical interface between the FPGA and memory has failed. Switch back to Standby and then Flash Access/Config mode to try reinitialising the memory.
8	Multistate Error	An error state has occurred where multiple processes has been started at once. Switch back to Standby and then Flash Access/Config mode to try reinitialising the memory.

Table 9-19: End of Life Information

EOL Value	EOL Information	Description
0	Not Defined	-
1	Normal	Normal level of consumption
2	Warning	Consumed +80% of reserved blocks
3	Urgent	Consumed +90% of reserved blocks

Table 9-20: Memory Usage Estimate

Usage Estimate Value	Usage Estimate
0	Not Defined
1	0% - 10%
2	10% - 20%
3	20% -30%
4	30% - 40%
5	40% - 50%
6	50% - 60%
7	60% -70%
8	70% - 80%
9	80% - 90%
10	90% - 100%
11	Exceeded max estimate

9.2.7.2 MEMORY_SIZE

This register contains the available memory size.

Register Address	0x0081
Default Value	0x00000000

Name	Bits	Access	Default	Description
Mem_Size	31-0	R	0	The available memory is chip dependent and is read from the flash memory unit on memory initialisation. The register value represents the available memory size in in packets (16380 bytes). E.g. 0x001E01E0 = 1966560 packets = 32212252800 bytes = 30 GB.

9.2.7.3 MEM_HLTH

These registers contains various metrics for evaluating the health of the onboard flash-based eMMC memory. The fields of these registers are inspired by traditional S.M.A.R.T. hard drive reports.

Register Address	0x0082 – 0x008A
Default Value	0x00000000

Name	Bits	Access	Default	Description
				MEM_HLTH_0 (0x0082)
Mem_Temp_0 [15:0]	15-0	R	0	eMMC controller temperature with a resolution of 0.25 °C.
Mem_Temp_1 [15:0]	31-16	R	0	eMMC NAND flash temperature with a resolution of 0.25 °C.
				MEM_HLTH_1 (0x0083)
Init_Bad_Block [15:0]	15-0	R	0	This value indicates the number of bad blocks detected after manufacturing of the flash memory (Secure Smart Report).
Run_Bad_Blocks [15:0]	31-16	R	0	This value indicates the number of bad blocks detected during the lifetime of the flash memory (Secure Smart Report).
				MEM_HLTH_2 (0x0084)
Rem_Spare_Blocks [15:0]	15-0	R	0	This value indicates the number of remaining spare blocks to be used to replace detected bad blocks (Secure Smart Report).
Pwr_Loss_Cnt [15:0]	31-16	R	0	This value indicates the number of times the memory has experienced sudden power loss without proper shutdown (Secure Smart Report).
				MEM_HLTH_3 (0x0085)
Init_Counter [31:0]	31-0	R	0	Cumulative initialization count of the memory (Secure Smart Report).
				MEM_HLTH_4 (0x0086)
Write_Count [31:0]	15-0	R	0	Total data written to memory with a resolution of 100MB (Secure Smart Report).
				MEM_HLTH_5 (0x0087)
Read_Count [31:0]	31-0	R	0	Total data read from memory with a resolution of 100MB (Secure Smart Report).
				MEM_HLTH_6 (0x0088)
Total_Uptime [31:0]	31-0	R	0	Total uptime of the memory in minutes (Secure Smart Report).
				MEM_HLTH_7 (0x0089)
Active_Uptime [31:0]	31-0	R	0	Total active uptime of the memory in minutes (Secure Smart Report).
				MEM_HLTH_8 (0x008A)
OverTemp_Uptime [15:0]	31-0	R	0	Total uptime of the memory during an overtem condition in minutes (Secure Smart Report).

9.2.7.4 MEM_WR_CTRL

This register controls the flash write process.

Register Address	0x0090
Default Value	0x00000000

Name	Bits	Access	Default	Description
Start	0	RW	0	This bit enables the flash memory write process. Any data received by the input of the memory interface block will be written into flash memory.
Stop	1	RW	0	This bit will stop the flash write process and add an end-of-file flag to the Packet ID. The current packet will be padded with the character 'A' (0x10) if the data does not end exactly on a packet boundary.
Abort	2	RW	0	This bit aborts the whole write process and should only be used if the stop control is unresponsive (Write operation hanging).
Reserved	31-3	R	0	-

9.2.7.5 MEM_WR_INDEX

Sets the flash memory write block index. Each block contains 4 bytes for a Packet ID and 16380 bytes of payload data.

Register Address	0x0091
Default Value	0x00000000

Name	Bits	Access	Default	Description
Index	31-0	RW	0	Sets the memory index where the write process will start. All 32 bits of the register may be written, but the maximum length is 23 bits. If the value that is written is larger than the available memory size the value will not be stored (The end boundary is indicated in the Section 9.2.7.2).

9.2.7.6 MEM_WR_LEN

Sets the number of bytes to write to memory. The number is a 64-bit number which is split into two registers.

Register Address	0x0092, 0x0093
Default Value	0x00000000

Name	Bits	Access	Default	Description
MEM_WR_LEN_0 (0x0092)				
Length_Lower	31-0	RW	0	This 32-bit value is the lower 32 bits of the 64-bit byte length value. If the write length (The full 64-bit value) is left as 0 the write process will write into memory until it is manually stopped or if it exceeds the memory limit.
MEM_WR_LEN_1 (0x0093)				
Length_Upper	31-0	RW	0	This 32-bit value is the upper 32 bits of the 64-bit byte length value. If the write length (The full 64-bit value) is left as 0 the write process will write into memory until it is manually stopped or if it exceeds the memory limit.

9.2.7.7 MEM_WR_STAT

This register shows the status of the memory write process.

Register Address	0x0094
Default Value	0x00000000

Name	Bits	Access	Default	Description
Active	0	R	0	This bit shows that the write process is active. The bit will stay active until the defined write process ends or it is stopped/aborted.
Complete	1	R	0	This bit shows that the memory write process is complete. Bit will clear when a new write process is started.
Error	2	R	0	This bit indicates an error has occurred during the write process. Bit will clear when a new write process is started.
Memory_OOB	3	R	0	This bit indicates the flash controller has tried to write past the end boundary of the flash memory (The end boundary is indicated in the Section 9.2.7.2). Bit will clear when a new write process is started.
Invalid_Index	4	R	0	This bit indicates that the index is not in the range of the available memory size detailed in Section 9.2.7.2. Writing a valid index will clear this bit.
Reserved	31-5	R	0	

9.2.7.8 MEM_WR_CNT

These registers shows how many bytes have been written in the current write transaction. This is a 64-bit number which is split into two registers. These registers are cleared upon a new memory write operation or a mode change to Standby mode.

Register Address	0x0095, 0x0096
Default Value	0x00000000

Name	Bits	Access	Default	Description
MEM_WR_CNT_0 (0x0095)				
Count_Lower	31-0	R	0	This 32-bit value is the lower 32 bits of the 64-bit byte count value.
MEM_WR_CNT_1 (0x0096)				
Count_Upper	31-0	R	0	This 32-bit value is the upper 32 bits of the 64-bit byte count value.

9.2.7.9 MEM_WR_PKT_CNT

This register contains the count of the number of packets written into memory in the current/previously active write process. This register is cleared upon a new memory write operation or a mode change to Standby mode.

Register Address	0x0097
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Packet_Cnt	31-0	R	0	The packet count indicates the number of packets written into memory (16380 bytes of data).

9.2.7.10 MEM_WR_ERR

This register contains the error information if an error occurs during the write process (For debugging).

Register Address	0x0098
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Err_Index [22:0]	22-0	R	0	The index value indicates at what memory index the write failed at.
Err_Timeout	23	R	0	Indicates if the error was a timeout error.
Err_SM [7:0]	31-24	R	0	State number where the error occurred in the internal write process state machine.

9.2.7.11 MEM_RD_CTRL

This register controls the memory-read process.

Register Address	0x00A0
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Start	0	RW	0	This bit enables the flash memory read process. The retransmission and read processes can run concurrently. The data is read from the memory and is sent to the crossbar to be routed to its desired destination.
Stop	1	RW	0	This bit will stop the flash read process.
Abort	2	RW	0	This bit aborts the whole read and retransmit process and should only be used if the stop control is unresponsive (Read/Retransmit operation hanging).
EOF_Stop	3	RW	0	If this bit is set to '1' the process will read from memory and if an EOF is detected in the Packet ID the read process will stop (Complete process).
Reserved	31-4	R	0	-

9.2.7.12 MEM_RD_INDEX

Sets the flash memory read block index. Each block contains 4 bytes for a Packet ID and 16380 bytes of payload data.

Register Address	0x00A1
Default Value	0x00000000

Name	Bits	Access	Default	Description
Index	31-0	RW	0	Sets the memory index where the read process will start. All 32 bits of the register may be written to, but only the top 23 bits are used. The block offset is limited to the available size of the flash memory. If the value that is written is larger than the available memory size the value will not be stored (The end boundary is indicated in the Section 9.2.7.2).

9.2.7.13 MEM_RD_LEN

Sets the number of packets to read from memory.

Register Address	0x00A2
Default Value	0x00000000

Name	Bits	Access	Default	Description
Length	31-0	RW	0	This 32-bit value is the number of packets to be read from the flash memory. If the read length is left as 0 the read process will read from memory until it is stopped, if it exceeds the memory limit or hits an EOF (EOF_Stop enabled).

9.2.7.14 MEM_RD_STAT

This register shows the status of the flash read process.

Register Address	0x00A3
Default Value	0x00000000

Name	Bits	Access	Default	Description
Active	0	R	0	This bit shows that the read process is active. The bit will stay active until the defined read process ends or it is stopped/aborted.
Complete	1	R	0	This bit shows that the flash read process is complete. Bit will clear when a new read process is started.
Error	2	R	0	This bit indicates an error occurred during the read process. Bit will clear when a new read process is started.
Memory_OOB	3	R	0	This bit indicates the flash controller has tried to read past the end boundary of the flash memory (The end boundary is indicated in the Section 9.2.7.2). Bit will clear when a new read process is started.
Invalid_Index	4	R	0	This bit indicates that the index is not in the range of the available memory size detailed in Section 9.2.7.2. Writing a valid index will clear this bit.
Reserved	31-5	R	0	-

9.2.7.15 MEM_RD_CNT

This register shows how many packets have been read.

Register Address	0x00A4
Default Value	0x00000000

Name	Bits	Access	Default	Description
Packet_Cnt	31-0	R	0	This value is the number of packets read. The value resets when a new read process is started.

9.2.7.16 MEM_RD_ERR

This register contains the error information if an error occurs during the read process (For debugging).

Register Address	0x00A5
Default Value	0x00000000

Name	Bits	Access	Default	Description
Err_Index [22:0]	22-0	R	0	The index value indicates at what memory index the read failed at.
Err_Timeout	23	R	0	Indicates if the error was a timeout error.
Err_SM [7:0]	31-24	R	0	State number where the error occurred in the internal read process state machine.

9.2.7.17 MEM_RT_CTRL

This register controls the retransmit function of the VestaLink S25.

Register Address	0x00B0
Default Value	0x00000000

Name	Bits	Access	Default	Description
Start	0	RW	0	This bit enables the retransmission process. The retransmission and read processes can run concurrently. The data is read from the memory and is sent to the crossbar to be routed to its desired destination.
Stop	1	RW	0	This bit instructs the VestaLink S25 to stop the retransmit process.
Priority	2	RW	0	This bit sets the retransmit mode priority. A value of '0' will instruct the VestaLink S25 to wait until the current memory read process has completed before sending the retransmit requests. A value of '1' will cause the VestaLink S25 to interrupt the read process after the current packet being processed has completed and start the available retransmit request (The read process will continue after queue is empty).
Clear_Queue	3	RW	0	Setting this bit will clear the retransmit request queue.
Reserved	31-4	R	0	-

9.2.7.18 MEM_RT_QUEUE

This register accepts the retransmit requests for the VestaLink S25. Writing to this register will inject the request into the retransmit queue, which can take up to 256 requests.

Register Address	0x00B1
Default Value	0x00000000

Name	Bits	Access	Default	Description
Index [23:0]	23-0	RW	0	This field specifies the Packet ID that is to be retransmitted. This is the index where the retransmit process will start reading from in memory. If the value that is written is larger than the available memory size the value will not be stored (The end boundary is indicated in the Section 9.2.7.2).
Length [7:0]	31-24	RW	0	This field specifies the number of consecutive packets to retransmit from the specified Packet ID. If a length of 0 is set the retransmit request will be ignored. The max length is 255. The max retransmit length in bytes is 4176900 bytes.

9.2.7.19 MEM_RT_STAT

This register contains the retransmit status fields.

Register Address	0x00B2
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Active	0	R	0	This bit shows that the retransmit process is active. The bit will stay active until the process is stopped.
Complete	1	R	0	This bit shows that the retransmit process is complete. The retransmit will remain active until manually stopped so complete bit is set when manually stopping an active retransmit process.
Error	2	R	0	This bit indicates an error occurred during the read process. It is cleared when a new retransmit process is started.
Memory_OOB	3	R	0	This bit indicates the flash controller has tried to read past the end boundary of the flash memory. (The end boundary is indicated in the Section 9.2.7.2). Bit will clear when a new retransmit process is started.
Invalid_Index	4	R	0	This bit indicates that the index is not in the range of the available memory size detailed in Section 9.2.7.2. Writing a valid index will clear this bit.
Queue_Full	5	R	0	This flag indicates that the retransmit FIFO is full.
Queue_Overflow	6	R	0	This flag indicates if the retransmit FIFO has overflowed.
Reserved	7	R	0	-
Queue_Count [7:0]	15-8	R	0	This field shows how many retransmit requests have been inserted into the retransmit FIFO. The FIFO is 255 entries deep.
Reserved	31-16	R	0	-

9.2.7.20 MEM_RT_CNT

This register shows how many retransmit packets have been read.

Register Address	0x00B4
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Packet_Cnt	31-0	R	0	This value is the number of retransmit packets read during current active retransmit session. The value resets when a new retransmit process is started.

9.2.7.21 MEM_RT_ERR

This register contains the error information if an error occurs during the retransmit process.

Register Address	0x00B5
Default Value	0x00000000

Name	Bits	Access	Default	Description
Err_Index [22:0]	22-0	R	0	The index value indicates at what memory index the retransmit failed at.
Err_Timeout	23	R	0	Indicates if the error was a timeout error.
Err_SM [7:0]	31-24	R	0	State number where the error occurred in the internal retransmit process state machine.

9.2.8 DATA ROUTING REGISTERS

9.2.8.1 TF_CTRL

Sets the operating modes for the transfer frame encapsulator.

Register Address	0x0110
Default Value	0x00000002

Name	Bits	Access	Default	Description
Enable	0	RW	0	This bit enables the transfer frame encapsulator. The framer is enabled when entering Config mode. If disabled the framer is bypassed and data is streamed directly into the selected encoder.
En_CRC	1	RW	1	The CRC16 generation can be enabled or disabled with this bit. CRC16 covers the entire transfer frame (Excluding ASM).
Idle_Mode [1:0]	3-2	RW	0	Sets the padding in the OID frame. (See Table 9-21).
Buf_Flush	4	RW	0	Setting this bit will flush any data stuck in the encapsulator buffer by injecting padding bytes into the transfer frame until a frame is filled. This bit is automatically cleared after the data is flushed.
Reserved	7-5	R	0	-
Padding [7:0]	15-8	RW	0xAA	This byte specifies the padding used when padding a frame or as OID data if OID padding is selected selected.
Reserved	31-5	R	0	-

Table 9-21: Idle Mode Padding

Value	Description
0x00	Pseudorandom Noise
0x01	Incrementing 8-bit counter
0x02	Padding with selected character (default ‘A’ (0x10))
0x03	Reserved

9.2.8.2 TF_BUF_CNT

This register contains the transfer frame encapsulator buffer.

Register Address	0x0111
Default Value	0x00000000

Name	Bits	Access	Default	Description
Buffer_Count [15:0]	15-0	R	0	The buffer count is the number of bytes written into the buffer of the encapsulator. A frame consists of the frame header and a data segment with a length of 16380 bytes. The framer waits until 16380 bytes is available in the buffer to then form a complete frame. If the buffer does not contain the necessary amount of data bytes the framer will generate an OID frame.
Reserved	31-16	R	0	-

9.2.8.3 TF_CONF

Sets the virtual channel IDs and spacecraft ID for the transfer frames.

Register Address	0x0112
Default Value	0x00002006

Name	Bits	Access	Default	Description
SPID [9:0]	9-0	RW	0x06	This field sets the spacecraft ID in the Transfer Frame Primary Header.
VCID [2:0]	12-10	RW	0x00	The VCID is the virtual channel number for the payload data in the Transfer Frame Primary Header. Take note that the OID frames are sent on channel 7.
VCID_RT [2:0]	15-13	RW	0x01	The VCID_RT is the virtual channel number for the retransmission payload data in the Transfer Frame Primary Header. Take note that the OID frames are sent on channel 7. **This field is only applicable when using DVB-S2 encoding. When using CCSDS encoding, retransmission payload data is sent on the configured VCID above.**
Reserved	31-16	R	0	-

9.2.8.4 TF_DATA_CNT

This register contains the count of data packets generated by the transfer frame encapsulator.

Register Address	0x0113
Default Value	0x00000000

Name	Bits	Access	Default	Description
Data_Count	31-0	R	0	The count is the number of data frames generated by the framer. The count is reset when the framer is bypassed.

9.2.8.5 TF_OID_CNT

This register contains the count of OID packets generated by the transfer frame encapsulator.

Register Address	0x0114
Default Value	0x00000000

Name	Bits	Access	Default	Description
Idle_Count	31-0	R	0	The count is the number of OID frames generated by the framer. The count is reset when the framer is bypassed.

9.2.8.6 TF_STREAM_CNT

This register contains the byte count of raw data that bypasses the transfer frame encapsulator.

Register Address	0x0115
Default Value	0x00000000

Name	Bits	Access	Default	Description
Stream_Count	31-0	R	0	When the framer is bypassed, the raw data is directly fed into the CCSDS / DVB-S2 core. This register value indicates the number of bytes that have bypassed the framer into the core. The counter is reset when the framer is enabled.

9.2.8.7 CROSSBAR

This register controls the data connections of the crossbar. The crossbar controls where the data flows inside the transmitter. The crossbar allows for a single data input to be connected to one or more data outputs (In case of multiple outputs connected to a single input the throughput rate regarding the data outputs will be determined by the slowest output). Inputs are connected to outputs by writing the input number into the desired output's register section. Writing a 0x00 to the output keeps it in a disconnected state and clears any held data in the crossbar. Unused inputs are kept in a disconnected state. The input values are given in Table 9-22. An example of the internal routings of the crossbar with a register value of 0x00001321 is shown in Figure 9-1.

Register Address	0x0120
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Test Pattern Checker [3:0]	3-0	RW	0	Output 1 - Writing a 4-bit input number to this register section connects the input to this output.
Transfer Framer [3:0]	7-4	RW	0	Output 2 - Writing a 4-bit input number to this register section connects the input to this output.
Memory Write [3:0]	11-8	RW	0	Output 3 - Writing a 4-bit input number to this register section connects the input to this output. Connecting memory read to memory write will put either operation into wait state as only one memory action can run at a time.
HS 0 Output [3:0]	15-12	RW	0	Output 4 - Writing a 4-bit input number to this register section connects the input to this output.
HS 1 Output [3:0]	19-16	RW	0	Output 5 - Writing a 4-bit input number to this register section connects the input to this output.
Reserved	23-20	R	0	-
Reserved	27-24	R	0	-
Reserved	31-28	R	0	-

Table 9-22: Input Values

Value	**Input Name**
0x01	Test Pattern Generator
0x02	HS 0 Input
0x03	HS 1 Input
0x04	SPI Input
0x05	Memory Read
0x0F-0x06	Reserved

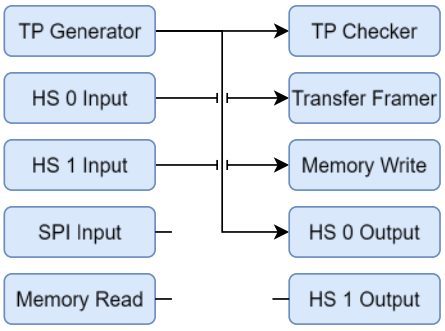


Figure 9-1: Example of crossbar internal connections.

9.2.8.8 TP_GEN_CTRL

This register controls the test pattern generator.

Register Address	0x0121
Default Value	0x00000000

Name	Bits	Access	Default	Description
Enable	0	RW	0	Enables the test pattern to be generated.
Pattern [1:0]	2-1	RW	0	Sets the type of pattern to be generated (Shown in Table 9-23).
EOF_Enable	3	RW	0	Determines if an EOF is attached to the last byte of the length defined test pattern (Length != 0). EOF is attached if bit is set to '1'.
Length [27:0]	31-4	RW	0x00	Sets the number of bytes (28-bit length) to be generated when the generator is enabled. Value of 0x00 continuously generates data while enabled.

Table 9-23: Test Pattern Type

Value	Test Pattern
0x00	8 Bit Counter (Incrementing)
0x01	PRBS7
0x03-0x02	Reserved

9.2.8.9 TP_GEN_CNT

This register contains the byte count of the generated test pattern.

Register Address	0x0122
Default Value	0x00000000

Name	Bits	Access	Default	Description
Gen_Count	31-0	R	0	This value indicates the number of bytes the test pattern has generated while enabled. The Value is reset when the generator is reenabled.

9.2.8.10 TP_GEN_RATE

This register contains the data rate that the generator is outputting the data.

Register Address	0x0123
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Data_Rate	31-0	R	0	The data rate is a running average of the number of bits generated in 1 second. The count is sampled every 1 second. If a constant rate is detected over 10 sample points the average will be set to the constant data rate value. The data rate is represented in this register as bits/second. A value of 0x09896800 is 160 Mbits/s.

9.2.8.11 TP_CHK_CTRL

This register controls the test pattern checker.

Register Address	0x0124
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Enable	0	RW	0	Enables the test pattern checker to start accepting data.
Pattern	2-1	RW	0	Sets the type of pattern to check for (Shown in Table 9-23).
Reserved	31-3	R	0	-

9.2.8.12 TP_CHK_STAT

This register contains the status of the test pattern checker.

Register Address	0x0125
Default Value	0x00000000

Name	**Bits**	**Access**	**Default**	**Description**
Pattern_Sync	0	R	0	This bit indicates that the checker pattern matches the incoming data and has synchronized to the data stream.
Reserved	15-1	R	0	-
Sync_Loss_Cnt [15:0]	31-16	R	0	The sync loss count value is incremented every time the pattern sync is lost.

9.2.8.13 TP_CHK_SYNC_CNT

This register contains the byte count of the incoming data into the test pattern checker.

Register Address	0x0126
Default Value	0x00000000

Name	Bits	Access	Default	Description
Sync_Count	31-0	R	0	This value indicates the number of bytes the test pattern checker has received while it is synced to a pattern.

9.2.8.14 TP_CHK_UNSYNC_CNT

This register contains the byte error count of the test pattern checker. If enough errors are received the check will lose sync.

Register Address	0x0127
Default Value	0x00000000

Name	Bits	Access	Default	Description
Unsync_Count	31-0	R	0	This value indicates the number of bytes the test pattern checker has received while it is not synced to a pattern (unsynced).

9.2.8.15 TP_CHK_RATE

This register contains the data rate that the test pattern checker is receiving the data at.

Register Address	0x0128
Default Value	0x00000000

Name	Bits	Access	Default	Description
Data_Rate	31-0	R	0	The data rate is a running average of the number of bits received in 1 second. The count is sampled every 1 second. If a constant rate is detected over 10 sample points the average will be set to the constant data rate value. The data rate is represented in this register as bits/second. A value of 0x09896800 is 160 Mbits/s.

9.2.8.16 TP_CHK_LST_CNT

This register contains the count of received data until an EOF is received.

Register Address	0x0129
Default Value	0x00000000

Name	Bits	Access	Default	Description
Last_Count	31-0	R	0	The last count is the count of bytes received up until a EOF is received. The count is only displayed/updated after an EOF is received.

9.2.9 PA STATUS AND CONTROL REGISTERS

9.2.9.1 PA_STAT

These registers show the status of the PA.

Register Address	0x0190, 0x0191
Default Value	0x00000000

Name	Bits	Access	Default	Description
				PA_STAT_0 (0x0190)
PA_On	0	R	0	This bit indicates the PA is powered on. This bit is read-only, to enable the PA, use the Mode register.
Over_Current	1	R	0	This flag is set when current through the PA exceeds the limit set in the PA_LIM register. When this happens, the PA will be switched off, and the VestaLink S25 will be returned to Config mode. The corresponding flag in the board status register will also be set.
Over_Temp [1:0]	3-2	R	0	The over temperature value indicates at which protection level the over temp issue has occurred.
Reserved	7-4	R	0	-
Measured_Power [7:0]	15-8	R	0	Measured output power. **SW < v2.6.0:** This value is reported with a 0.1 dBm resolution, starting at 20 dBm. The measured power in dBm can thus be obtained by making use of the equation below: $$P_{measured}[dBm] = \frac{Measured_Power[7:0]}{10} + 20$$ **SW ≥ v2.6.0:** This value is reported with a 0.25 dBm resolution. The measured power in dBm can thus be obtained by making use of the equation below: $$P_{measured}[dBm] = \frac{Measured_Power[7:0]}{4}$$
PA_Temp [15:0]	31-16	R	0	Measured temperature of the PA. This is a signed 16-bit number, reporting temperature in 0.25°C increments.
				PA_STAT_1 (0x0191)
Current	15-0	R	0	Measured current through the PA. Reported in 0.1 mA increments.
PowerQuanta	31-16	R	0	The quanta value reported by the power detector ADC.

9.2.9.2 PA_CONF

These registers control the PA configuration, including target power, PID mode, and over-temperature behaviour

Register Address	0x0192, 0x0193
Default Value	-

Name	Bits	Access	Default	Description
				PA_CONF_0 (0x0192)
Target_Power	7-0	RW	-	Set the target output power in dBm. Valid values are 14-20 for Flatsats and 27-33 for evaluation/flight models. This value is valid(used) when the PID Mode field in PA_Conf_1 is not set to off.
Reserved	31-8	R	0	-
				PA_CONF_1 (0x0193)
PID_Mode	7-0	RW	0x01	Control mode for the PA. Table 9-24 shows possible values for this field. The default setting is set to Power Detector with Temperature.
Disabled_Over_Temp	8	RW	0	This bit disables the over temperature protection for the PA.
Reserved	31-9	R	0	-

Table 9-24: Possible values for PID Mode field

Mode	Value
Off	0x00
Power Detector with Temperature	0x01
Power Detector only	0x02
Open-Loop with Temperature	0x03
Open-loop	0x04

9.2.9.3 PA_LIM

This register sets the current and temperature warning limit of the PA.

Register Address	0x0197
Default Value	0x005A01C2

Name	Bits	Access	Default	Description
Current_Limit	15-0	RW	0x01C2	Sets the current limit for this PA. If the PA exceeds this limit, the PA will be turned off, and the OverCurrent flag in the PA_Status register will be set. The over current trip is implemented on a hardware level.
Temp_Warn	31-16	RW	0x005A	The temperature limit that will trigger temperature warning if the PA temperature exceeds this limit.

9.2.9.4 PA_TIMEOUT

The PA on-time limit of the board. The default timeout is set 18 minutes.

Register Address	0x0198
Default Value	0x04380000

Name	Bits	Access	Default	Description
Current_Uptime [15:0]	15-0	R	0x0000	The uptime indicates the number of seconds the PA has been on.
Uptime_Lim [15:0]	31-16	RW	0x0438	The time limit that will trigger the board to go back into Standby mode. The time limit is set in seconds. Set the value to 0x0000 to ignore the timeout procedure.

10 APPENDIX B: TYPICAL POWER CONSUMPTION

The VestaLink S25 has various operating modes which have different power consumption profiles. The main factors determining the power consumption is:

- Mode of operation.
- RF power setting of the PA when transmitting RF.
- Transmitter operating temperature.

The VestaLink S25 has a single power input interface (V_{BAT} pin on the PC104 connector). Refer to "ICD-01-10022" for details. The input can run from 16V and was used to generate Table 10-1.

Table 10-1: Typical Power consumption.

Mode / RF Power (dBm)	DC Power (W)
DISABLED	0.2
STANDBY	1.4
FLASH ACCESS (IDLE)	2
FLASH ACCESS (READ/WRITE)	2.2
CONFIG	5.6
TRANSMIT 27	10
TRANSMIT 28	10.5
TRANSMIT 29	11
TRANSMIT 30	11.5
TRANSMIT 31	12.2
TRANSMIT 32	13
TRANSMIT 33	14

The VestaLink S25 uses switching regulators and thus the current draw is dependent on the input voltage supply.

11 APPENDIX C: RF PERFORMANCE

The RF performance of a transmitter can be measured using the following modulation parameters:

- RF power output
- Error vector magnitude (EVM)
- Occupied bandwidth (99%)
- Emission mask compliance (IRU-R SM1541 recommendations)

All resuls listed below were taken at 25°C.

The VestaLink S25 is calibrated to have an EVM of less than 7% across all frequency and temperature combinations.

11.1 DVB-S2

Table 11-1: Typical RF Performance at 5 Msps (DVB-S2).

Modulation	RF Power (dBm)	EVM (% RMS)	Emission Mask	Occupied bandwidth (MHz)
QPSK	33	<7	Pass	<6
8PSK	33	<7	Pass	<6
16APSK	33	<7	Pass	<6
32APSK	33	<7	Pass	<6

11.1.1 EVM



Figure 11-1: Typical QPSK EVM at 2250 MHz with 33 dBm RF power (DVB-S2).



Figure 11-2: Typical 8PSK EVM at 2250 MHz with 33 dBm RF power (DVB-S2).

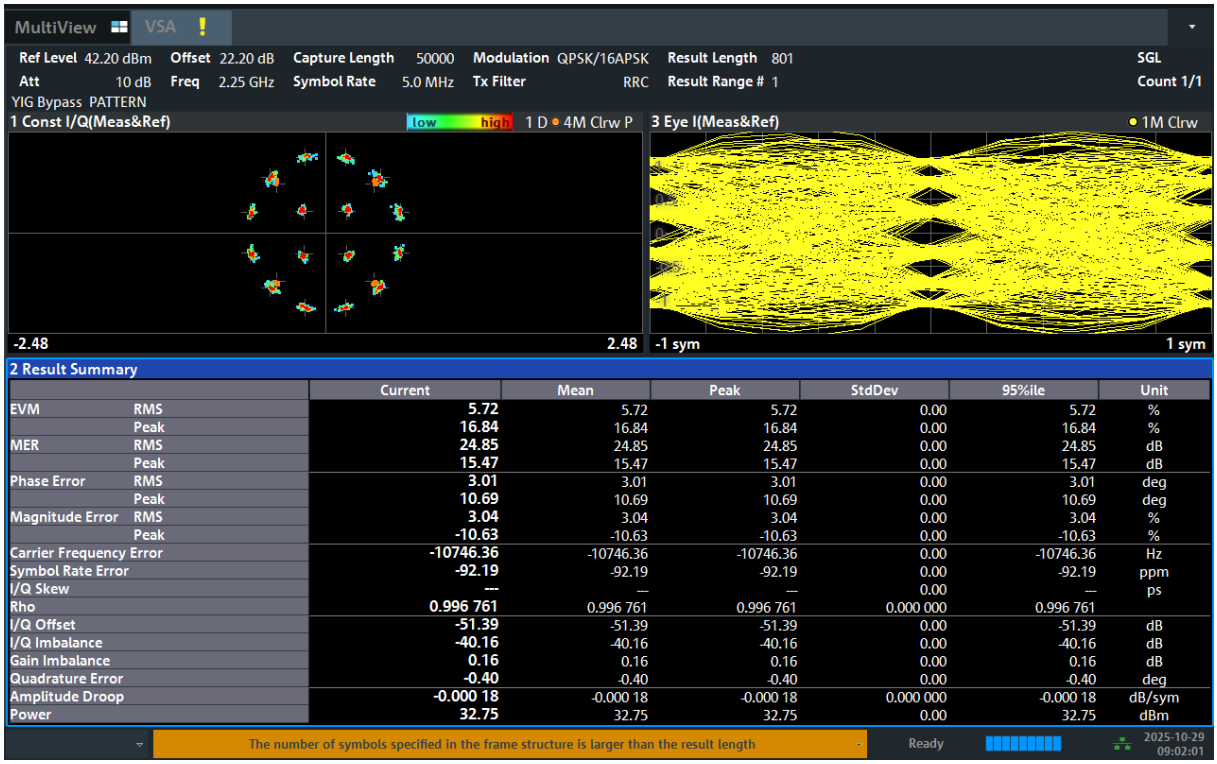


Figure 11-3: Typical 16APSK EVM at 2250 MHz with 33 dBm RF power (DVB-S2).

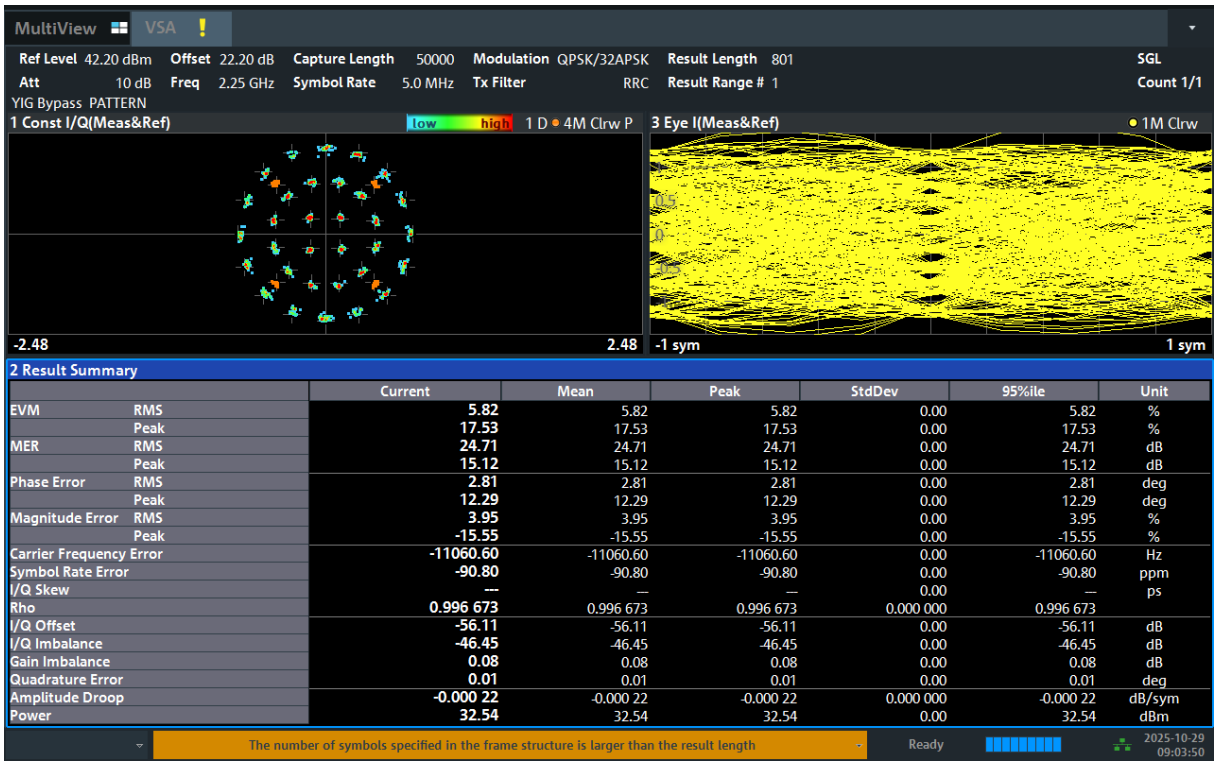


Figure 11-4: Typical 32-APSK EVM at 2250 MHz with 33 dBm RF power (DVB-S2).

11.1.2 EMISSION MASK

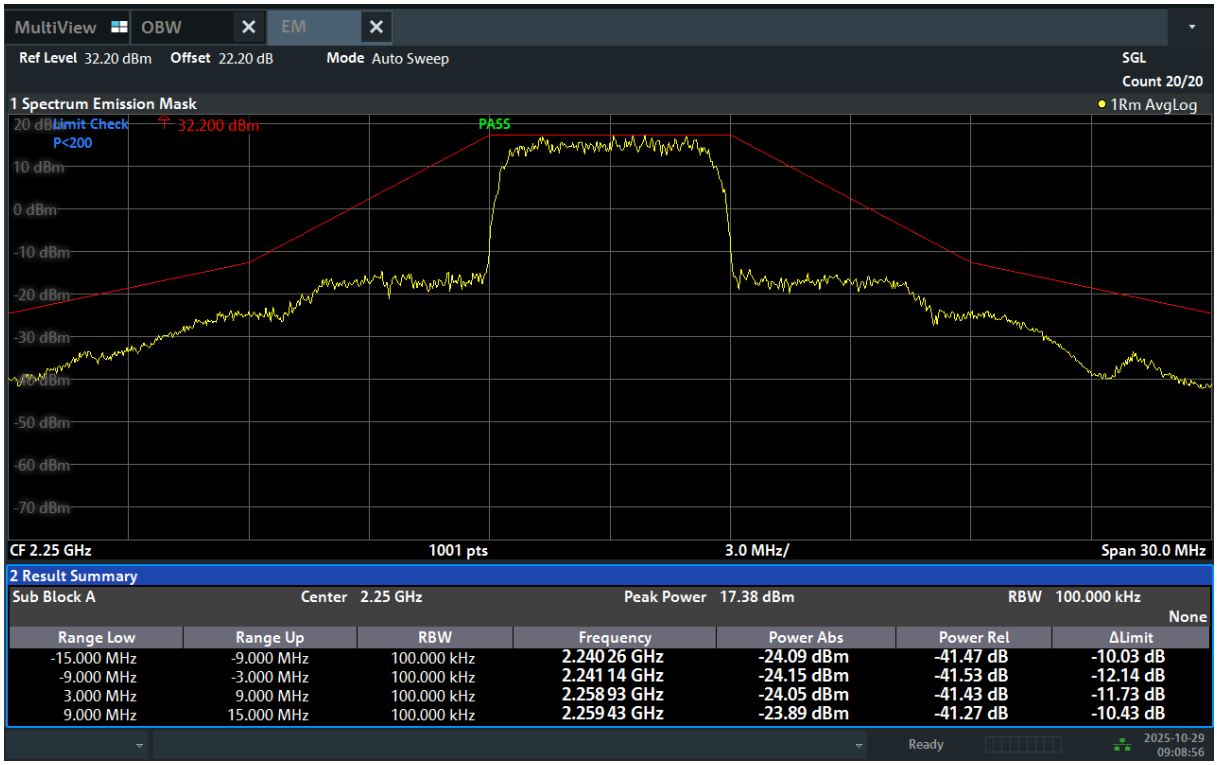


Figure 11-5: Typical QPSK emission mask at 2250 MHz with 33 dBm RF power (DVB-S2).

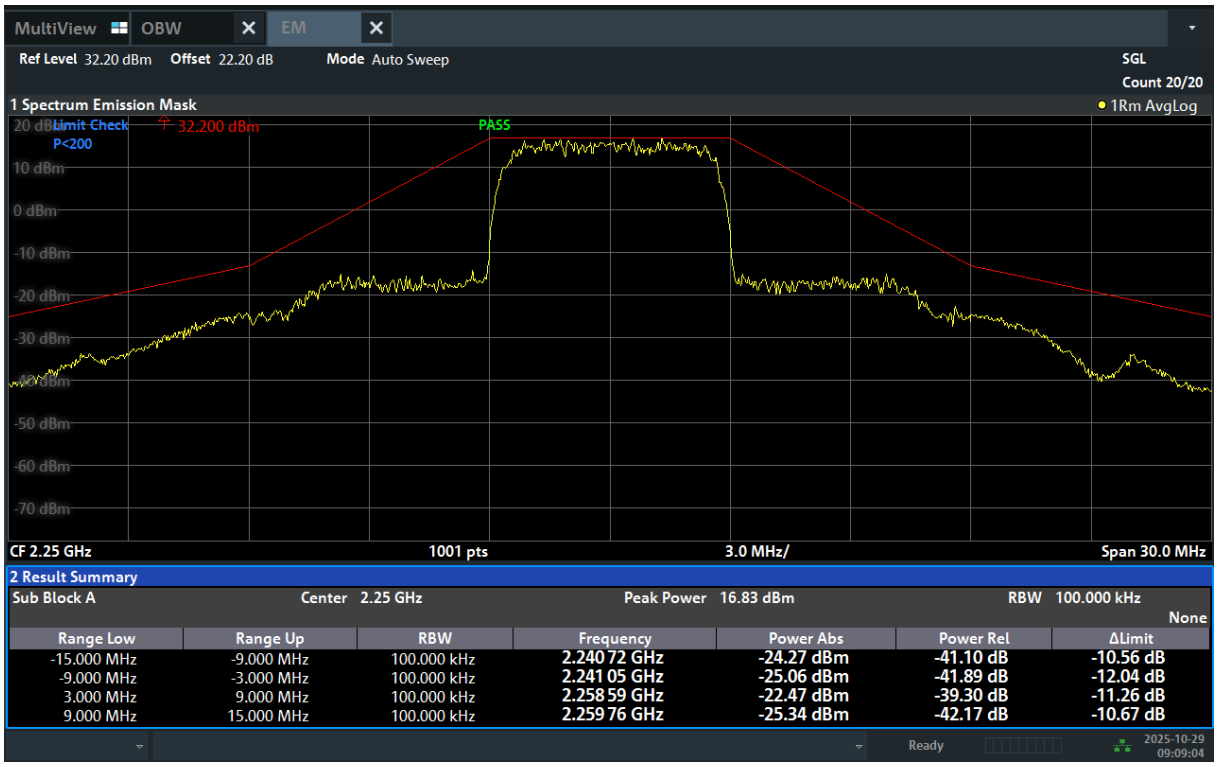


Figure 11-6: Typical 8PSK emission mask at 2250 MHz with 33 dBm RF power (DVB-S2).

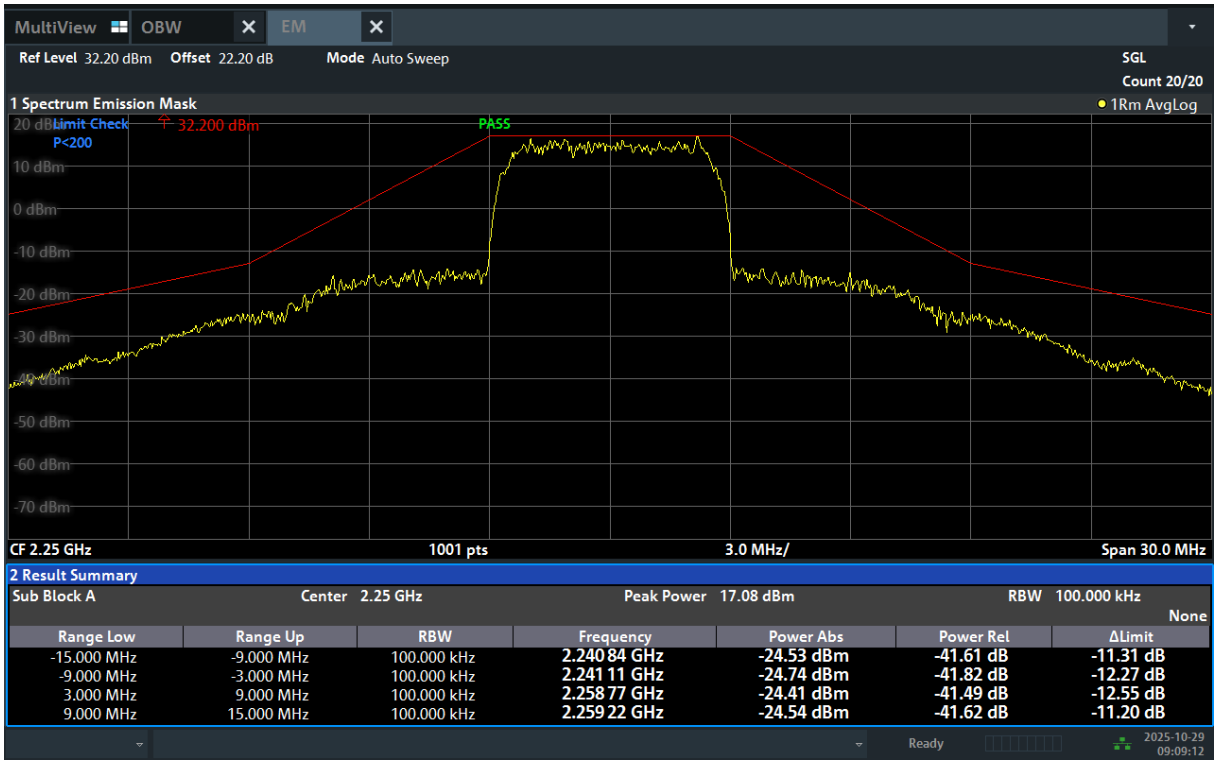


Figure 11-7: Typica 16APSK emission mask at 2250 MHz with 33 dBm RF power (DVB-S2).

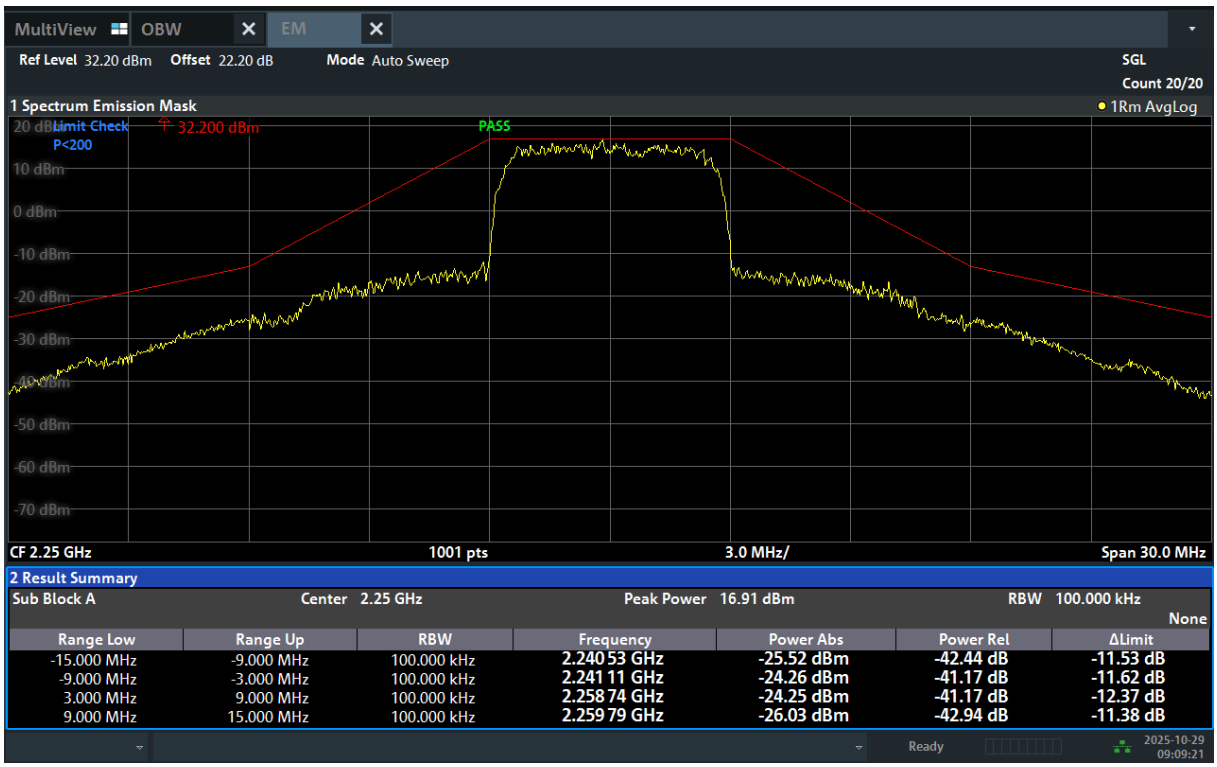


Figure 11-8: Typical 32APSK emission mask at 2250 MHz with 33 dBm RF power (DVB-S2).

11.1.3 OCCUPIED BANDWIDTH

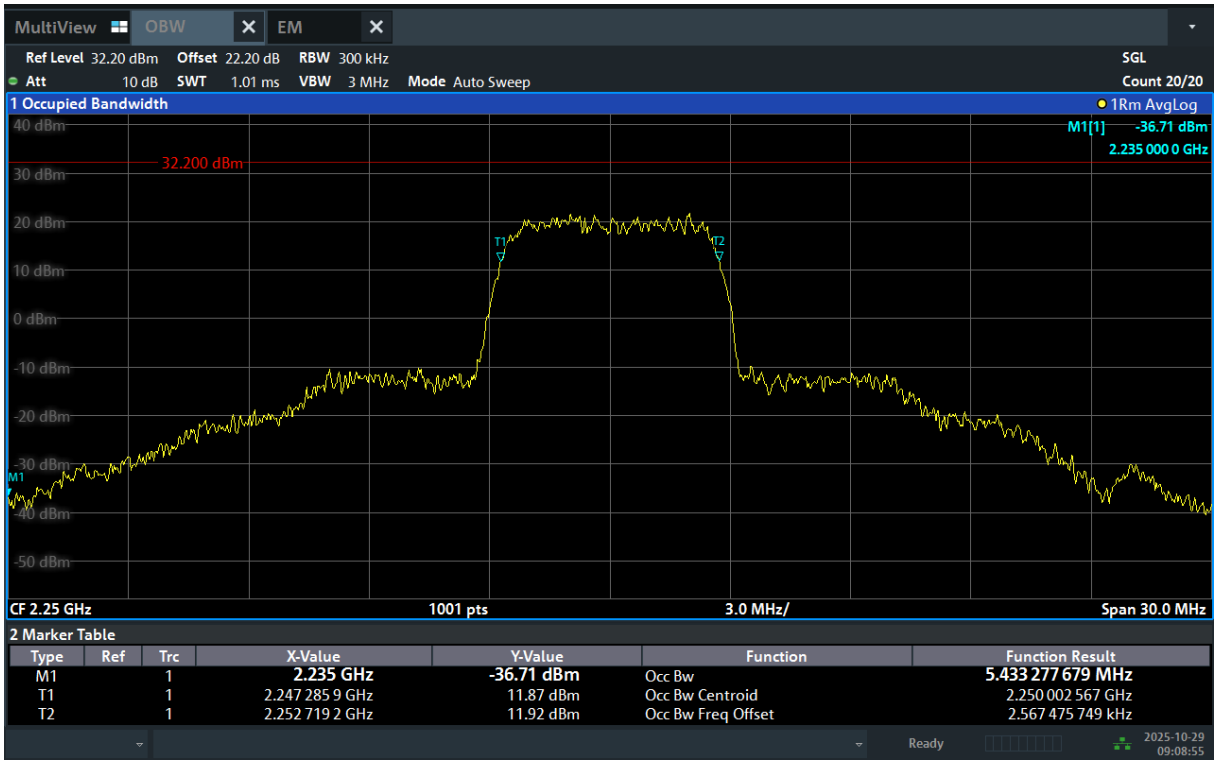


Figure 11-9: Typical QPSK 99% occupied bandwidth at 2250 MHz with 33 dBm RF power (DVB-S2).

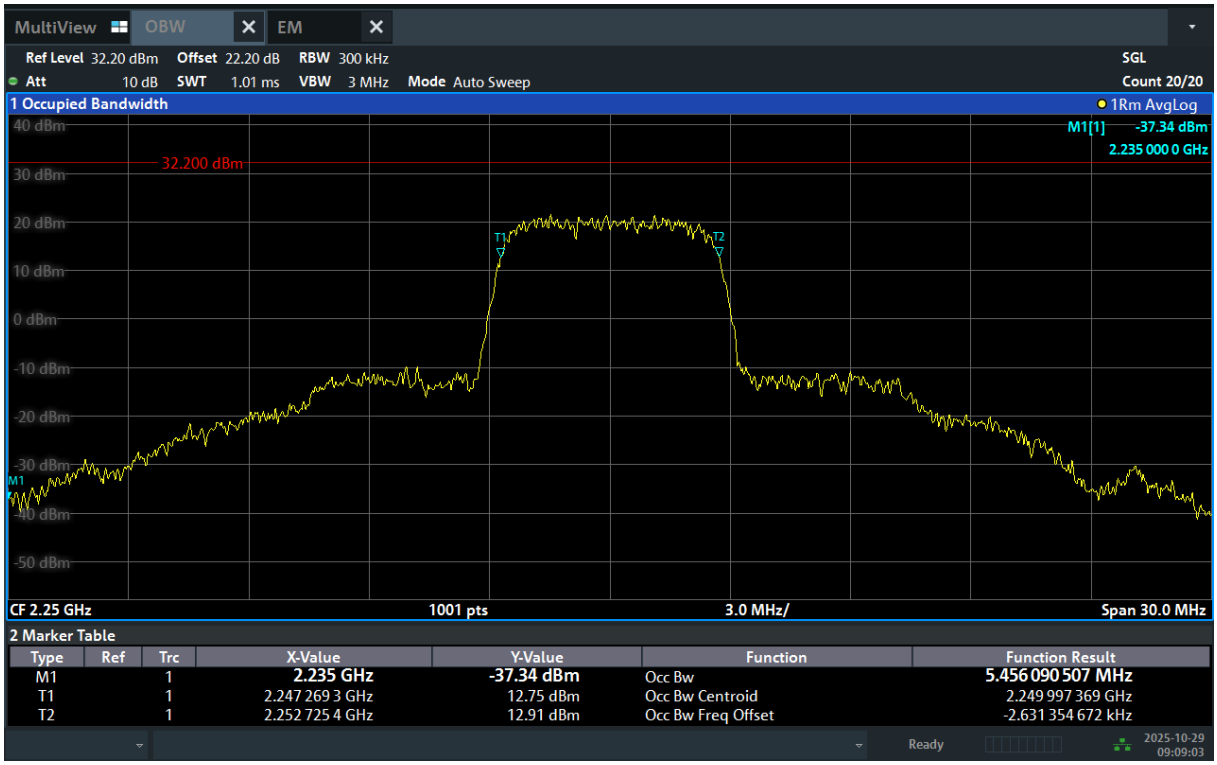


Figure 11-10: Typical 8PSK 99% occupied bandwidth at 2250 MHz with 33 dBm RF power (DVB-S2).

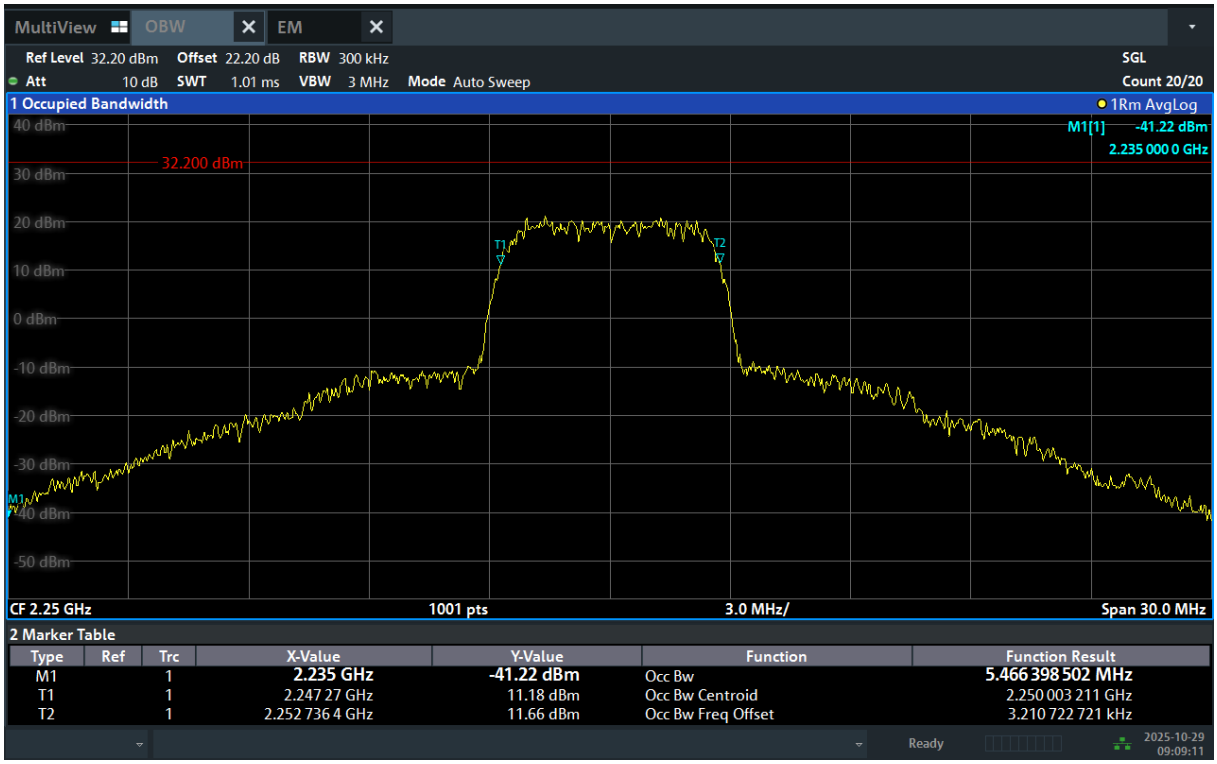


Figure 11-11: Typical 16APSK 99% occupied bandwidth at 2250 MHz with 33 dBm RF power (DVB-S2).

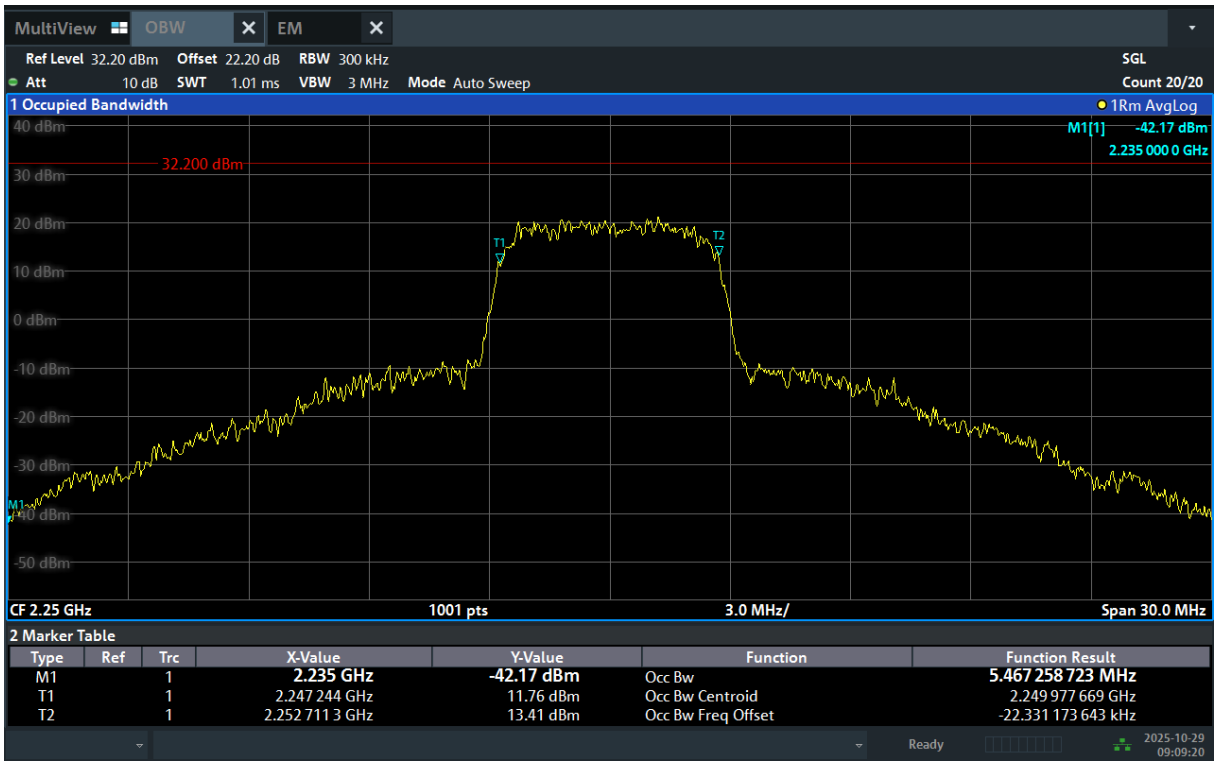


Figure 11-12: Typical 32APSK 99% occupied bandwidth at 2250 MHz with 33 dBm RF power (DVB-S2).

12 APPENDIX D: EXAMPLE SCENARIOS

This section lists example scenarios with reproducible steps for the VestaLink S25. This section is primarily divided into the different transmit modes and then secondarily divided into the specific example scenarios. The data interface configurations are predominantly performed in Flash Access mode while data transmission configurations are predominantly performed in Transmit mode. Refer to Section 8.2.1 for the different transmit modes and limitations.

12.1 FLASH ACCESS MODE CONFIGURATION

The following scenarios consist of interaction with different data interfaces in Flash Access Mode. The typical Flash Access mode configuration follows the sequence in Figure 12-1:

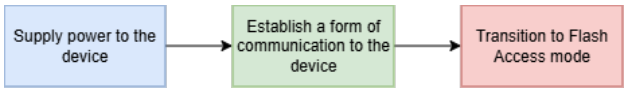


Figure 12-1: Typical Flash Access mode configuration process

To transition the device into Flash Access Mode, follow the sequence below:

Objective: Transition the VestaLink S25 to Flash Access Mode

Step	Register Name	Operation	Address	Target Value	Description	Reference
1	Mode	Write	0x0060	0x00000001	Set bits Mode [7:0] to 0x01 (Flash Access)	9.2.5.1

12.1.1 LVDS CONFIGURATION

The following scenarios describe the register sequences to interface with various LVDS receive configurations. The typical LVDS configuration and validation process can be divided into the sequence that follows in Figure 12-2:

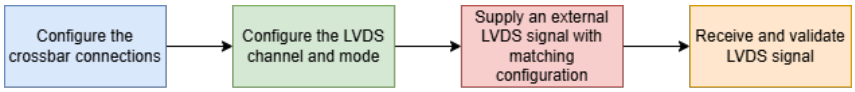


Figure 12-2: Typical LVDS configuration and validation process

12.1.1.1 HS NODE 0 & SINGLE CHANNEL SDR CONFIGURATION

This configuration validates the data path for a Single Data Rate (SDR) signal using the primary high-speed interface (HS node 0).

Objective: Configure the VestaLink S25 to capture a valid SDR LVDS signal on rising-edge latches via HS0. The test signal must originate from an external data source with the same LVDS configuration and 8-bit counter pattern.

Step	Register Name	Operation	Address	Target Value	Description	Reference
1	Crossbar	Write	0x0120	0x00000002	Set bits Test Pattern Checker [3:0] to 0x2 (HS0).	9.2.8.7
2	TP_Chk_Ctrl	Write	0x0124	0x00000001	Set bit Enable [0] to 1 and Pattern [2:1] to 0 (8 Bit Counter).	9.2.8.11
3	HS_Ctrl	Write	0x0021	0x00000001	Set bit Enable [0] to 1, Channel [2:1] to 0 (Only D0), and Mode [4:3] to 0 (SDR with Rising edge latching).	9.2.3.2
4	TP_Chk_Sync_Cnt	Read	0x0126	-	Read the number of bytes synced to a pattern. This value should increment while a valid LVDS signal and pattern is received.	9.2.8.13
5	TP_Chk_Unsync_Cnt	Read	0x0127	-	Read the number of bytes not synced to a pattern. This value should not increment while a valid LVDS signal and pattern is received.	9.2.8.14

Note: Ensure that the LVDS clock speed is maintained within limits dictated by the ICD Document (ICD-01-10022).

12.1.1.2 HS NODE 1 & DUAL CHANNEL DDR CONFIGURATION

This configuration validates the data path for a Dual Data Rate (SDR) signal using the secondary high-speed interface (HS node 1).

Objective: Configure the VestaLink S25 to capture a valid DDR LVDS signal via HS1. The test signal must originate from an external data source with the same LVDS configuration and 8-bit counter pattern.

Step	Register Name	Operation	Address	Target Value	Description	Reference
1	Crossbar	Write	0x0120	0x00000003	Set bits Test Pattern Checker [3:0] to 0x3 (HS1).	9.2.8.7
2	TP_Chk_Ctrl	Write	0x0124	0x00000001	Set bits Enable [0] to 1 and Pattern [2:1] to 0 (8-bit counter).	9.2.8.11
3	HS_Ctrl	Write	0x0021	0x00000014	Set bits Enable [0] to 1, Channel [2:1] to 2 (D0 & D1), and Mode [4:3] to 2 (DDR).	9.2.3.2
4	TP_Chk_Sync_Cnt	Read	0x0126	-	Read the number of bytes synced to a pattern. This value should increment while a valid LVDS signal and pattern is received.	9.2.8.13
5	TP_Chk_Unsync_Cnt	Read	0x0127	-	Read the number of bytes not synced to a pattern. This value should not increment while a valid LVDS signal and pattern is received.	9.2.8.14

Note: Ensure that the LVDS clock speed is maintained within limits dictated by the ICD Document (ICD-01-10022).

12.1.2 WRITING TO MEMORY

The following scenarios describe the register sequences to write to the VestaLink S25's memory. The typical Memory Write configuration and validation process can be divided into the sequence that follows in Figure 12-3:

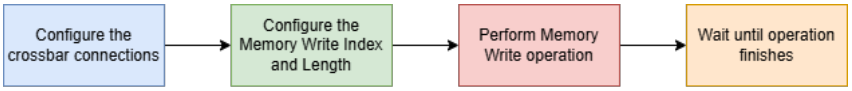


Figure 12-3: Typical Memory Write configuration process

12.1.2.1 TEST PATTERN CONFIGURATION

This configuration validates the data path for a memory write operation using a test pattern.

Objective: Configure the VestaLink S25 to write 8192 bytes of an 8-bit counter test pattern to memory index 50.

Step	Register Name	Operation	Address	Target Value	Description	Reference
1	Crossbar	Write	0x0120	0x00000100	Set bits Memory Write [11:8] to 0x1 (Test Pattern Generator).	9.2.8.7
2	TP_Gen_Ctrl	Write	0x0121	0x00000001	Set bits Enable [0] to 1 and Pattern [2:1] to 0 (8-bit counter)	9.2.8.8
3	Mem_Wr_Index	Write	0x0091	0x00000032	Set bits Index [31:0] to 0x32 (Index 50)	9.2.7.5
4	Mem_Wr_Len_0	Write	0x0092	0x00002000	Set bits Length_Lower [31:0] to 0x32 (8192 bytes total length)	9.2.7.6
5	Mem_Wr_Len_1	Write	0x0093	0x00000000	Set bits Length_Upper [31:0] to 0x00 (8192 bytes total length)	9.2.7.6
6	Mem_Wr_Ctrl	Write	0x0090	0x00000001	Set bits Start [0] to 1 (start write operation)	9.2.7.4

12.1.3 READING FROM MEMORY

The following scenarios describe the register sequences to read from the VestaLink S25's memory. The typical Memory Read configuration and validation process can be divided into the sequence that follows in Figure 12-4:

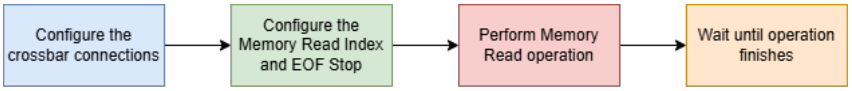


Figure 12-4: Typical Memory Write configuration process

12.1.3.1 TEST PATTERN CONFIGURATION

This configuration validates the data path for a memory read operation with a test pattern. This section is to follow directly after successful completion of the memory write operation performed in Section 12.1.2.1.

Objective: Configure the VestaLink S25 to read bytes of an 8-bit counter test pattern from memory index 50 until EOF is reached.

Step	Register Name	Operation	Address	Target Value	Description	Reference
1	Crossbar	Write	0x0120	0x00000005	Set bits Test Pattern Checker [3:0] to 0x5 (Memory Read).	9.2.8.7
2	TP_Chk_Ctrl	Write	0x0124	0x00000001	Set bits Enable [0] to 1 and Pattern [2:1] to 0 (8-bit counter).	9.2.8.11
3	Mem_Rd_Index	Write	0x00A1	0x00000032	Set bits Index [31:0] to 0x32 (Index 50)	9.2.7.12
4	Mem_Rd_Ctrl	Write	0x00A0	0x00000009	Set bits Start [0] to 1 and EOF_Stop [3] to 1	9.2.7.11
Arbitrary wait time (e.g. 500 ms)						
5	TP_Chk_Sync_Cnt	Read	0x0126	-	Read the number of bytes synced to a pattern. This value should be 8192 after the memory read has finished.	9.2.8.13
6	TP_Chk_Unsync_Cnt	Read	0x0127	-	Read the number of bytes not synced to a pattern. This value should be 8188 after the memory read has finished.	9.2.8.14

Note: The arbitrary wait time is dependent on the length of data that is required to be read from memory. A memory read is valid provided that the TP_Chk_Sync_Cnt aligns with the length of data written into memory.

13 DOCUMENT CONTROL AND APPROVAL

Rev	Description of Change	Author	Approved By	Date
A1	Original	S Pretorius	T Heidemann	2023/10/23
A2	Fixed references to sub-sections.	S Pretorius	T Heidemann	2023/11/06
A3	Updated register map.	S Pretorius	T Heidemann	2024/04/12
A4	Added Crossbar and Dual HS Interface (Updated register map).	S Pretorius	T Heidemann	2024/09/04
A5	Updated document formatting.	S Pretorius	T Heidemann	2024/10/21
A6	Fixed register map. Moved to new template.	S Pretorius	T Heidemann	2025/06/24
A7	Usermanual data restructure (Overall update)	S Pretorius	T Heidemann	2025/08/01
A8	Document restructuring. Added Protocol field to CAN_CONF register & added CONF & STAT registers for CAN1. CAN_CSP_CONF register added. CANTS_CONF register added. Various MEM_HLTH registers added. Added clarification on when MEM_WR_CNT and MEM_WR_PKT_CNT registers are cleared. Measured_Power field for PA_STAT_0 register range and resolution updated (SW v2.6.0). Updated typical power usage. Updated X_Drv_Amp_5V to X_Drv_Amp_3V3. Added emmission masks for all DVB-S2 MODCODs. Added RF performance section for CCSDS. Updated emission mask specification from 'SFCG 21-2R4' to 'ITU-R SM 1541'. Added "Example Scenarios" section. “RF Performance” appendix updated. Product Name Update	S Pretorius A vd Westhuizen L Brittz N Langeveldt		2026/06/25